



Adoption of low-code and no-code development: A systematic literature review and future research agenda

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ABSTRACT

Context: Low-code/no-code (LCNC) is an emerging technology trend that extends software development beyond professional software engineers, making it accessible to individuals throughout organizations and society.

Objective: We aim to provide a systematic review of the current research on the adoption of LCNC technologies within citizen development (CD) practices for digital transformation (DT), and to propose a research agenda for this field.

Method: This review is primarily conducted using a multi-phase systematic literature review of publications from the past five years, i.e., between 2017 and 2023.

Results: We identified 40 primary studies that describes the application of LCNC development and CD practices, the theoretical lenses/frameworks used, and the associated benefits and challenges.

Conclusion: In this study, we present three key contributions. First, we provide a comprehensive review of the benefits, challenges, theoretical perspectives, and methods used to explore LCNC and CD adoption. Second, we introduce a framework designed to guide managers in effectively adopting LCNC and CD practices. Finally, our systematic review uncovers gaps in existing research and identifies opportunities for further exploration, which paves the way for a future research agenda.

1. Introduction

The rapid emergence of digital technologies and platforms is transforming the software development landscape which has enabled unprecedented accessibility and innovation across organizations and society (Carroll et al., 2024). The growing reliance on digital technologies highlights its dual role as a transformative change and a tool to address some complex challenges across various domains. For example, in recent times, the use of digital technologies may be motivated by the need to survive and gain sovereignty against oppression (Qureshi, 2022), combat pandemic crises (He et al., 2021), improve business efficiency and effectiveness (Magesa and Jonathan, 2021), and participation for farmers in the agriculture value chain (Smidt and Jokonya, 2022); the role of information and communication technologies (ICT) are significant for modern society. For modern organizations, digital development (i.e., the process of creating, implementing, and advancing digital technologies and solutions) is fundamental to delivering digital transformation (DT) as a gateway to transforming businesses (Magesa and Jonathan, 2021). DT starts at the business level within

organizations, driving transformative changes in operations. These changes include migrating paper-based documents to digital information systems, integrating existing digital and manual process automation systems, and completely redesigning a business model to enhance operational efficiency and effectiveness (Metrolho et al., 2019). The goal of DT may be described as creating fundamental change in business value by integrating and leveraging digital technologies for business solutions. However, technological motivations often prioritize automating manual operations, such as workflows or tasks (e.g., invoice processing). Automating manual operations increases productivity, enhances agility and operational efficiency, enhances product and service innovations, and customer interactions. In addition to lowering labour and resource costs in daily business operations, it also increases revenue streams for organizations (Kane et al., 2015; Matt et al., 2015; Vial, 2019). The need for DT is becoming more important due to the growing expectations for more skilful engineers (Lara-Prieto and Flores-Garza, 2021), automatic coding (Woo, 2020), pressure to reduce budget costs (Fernandes et al., 2020), shorter development cycles, and greater alignment between business and IT departments (Elshan et al.,

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2023; Iho et al., 2021; Khorram et al., 2020; Ullrich et al., 2021).

However, the need for digital leadership (Magesa and Jonathan, 2021), the shortage of software developers (Carroll et al. 2021), and the demand for improved business operational efficiency, along with other priorities, have been regarded as essential to addressing the challenges contributing to DT failures in many organizations (Andriole, 2020; Kane et al., 2015; Lebens et al., 2021; Tekic and Koroteev, 2019). In addressing these challenges, some researchers (e.g., Carroll et al., 2021) and practitioners such as Gartner (Hurlburt, 2021) have started to recognize the business value of using disruptive low-code/no-code (LCNC) software technologies, driven by citizen development (CD), as a viable solution for leading DT initiatives in organizations (Carroll et al., 2021; Lebens et al., 2021; Sanchis et al., 2020; Yan, 2021). LCNC refers to technologies that allow for the development of software applications with little or no coding expertise, whereas CD is a process and practice of empowering non-technical employees to access and use LCNC technologies to develop software applications (Carroll and Maher, 2023). According to Gartner (2021), the LCNC development technologies market is projected to grow beyond the 23% recorded in 2021 (estimated to be \$13.8 billion) to around 30% since the end of 2022. It is also projected that by 2024, LCNC will represent 75% of software application development (Hurlburt, 2021). Some LCNC tools require little or no coding, are developed using engineering methods such as visual programming, machine learning, and block modelling languages to efficiently manage the development and deployment of software applications throughout the development cycle (Alsaadi et al., 2021; Bexiga et al., 2020; Palomés et al., 2021). In general, the rise of LCNC development platforms such as OutSystems, Microsoft PowerApps, Salesforce, Appian, Mendix and Zoho Creator are deemed to be beneficial for both people with and without coding knowledge and experience in software development (Alsaadi et al., 2021).

Research suggests that non-IT employees or business professionals without coding or technical skills (i.e., citizen developers) often derive greater benefits from using LCNC technology to develop software, compared to traditional software developers (Carroll and Maher, 2023). By utilizing these digital technologies to automate the development and deployment of software applications for addressing specific problems; they are now playing a critical transformative role as citizen developers in driving DT initiatives within organizations (Carroll, et al., 2023; Lebens et al., 2021; Palomés et al., 2021). The benefits of CD and LCNC include improving operational efficiency and effectiveness (Bexiga et al., 2020; Pacheco et al., 2021), empowering employees (Alsaadi et al., 2021; Hoogsteen and Borgman, 2022), reducing IT costs (Schenkenfelder et al., 2023), and better alignment between business and IT units (Elshan et al., 2023).

Despite the hype surrounding the use of LCNC technologies for piloting software development applications for business solutions (Carroll et al., 2021; Corral et al., 2021); studies suggest that these technologies are largely underutilized in terms of adoption (Bhattacharyya and Kumar, 2021), and their technical impact on business improvement remains questionable (Cabot, 2020). This uncertainty leads to a mix of cautious optimism and criticism, along with other challenges (e.g., security, cultural change) regarding their adoption within organizations (Michael and Wortmann, 2021; Palomés et al., 2021). Therefore, this research aims to address the following research questions:

- *RQ1 – What are the benefits and challenges associated with the adoption of LCNC technologies and CD practices in organizations?* The use of LCNC technologies in software development practices holds promise for business and technical professionals, yet they remain relatively new areas for research and industry application. This question explores the state of research on the theoretical, methodological, and practical contributions to studying the adoption of LCNC and CD adoption to drive DT initiatives within organizations.

- *RQ2 – How do organizations evaluate the methods they use in measuring the benefits of LCNC adoption and CD practices?* As an emerging business trend, this question offers insights into the way practitioners are measuring or evaluating the evidence of the benefits derived from LCNC adoption, by exploring and identifying relevant empirical studies for practical application.

Answering these questions deepens our understanding of the inherent benefits and challenges that are associated with socio-technical perspectives of LCNC technologies and CD practices for employees work and their organizations. In answering our research questions, we identified the need to undertake a multi-phase systematic literature review to better understand the benefits, challenges, and research opportunities inherent in CD for DT (Luo et al., 2021; Overeem and Jansen, 2021). However, we found a multi-vocal systematic review by Bucaioni et al. (2022) with a comparative focus on low-code development and model-driven engineering. Similar conceptual literature reviews include Berardi et al. (2023), Binzer & Winkler (2022), Curty et al. (2022), Sufi (2023). Nevertheless, our multi-phase systematic literature review approach by Wolfswinkel et al. (2013) builds on their research contributions regarding low-code adoption, and provides a deeper understanding of its associated benefits and challenges, ultimately helping to establish a research agenda.

Furthermore, the democratization of technology was not the result of a single company or industry, but rather driven by a cultural revolution where LCNC development platforms continue to emerge and adapt to human needs and digital skillset gaps (Elshan et al., 2023). Recent studies have highlighted the growing demand for more innovative and transformative use of ICT for social, economic, and environmental development (Roztocki et al., 2019; Qureshi, 2022; Zheng et al., 2018). ICT for development research seeks to understand the rationale for developing technologies, the intended users, and how these technologies become embedded or operationalized within organizations (Roztocki et al., 2019; Zheng et al., 2018). This systematic review contributes to this understanding by exploring and retrieving relevant studies on the adoption of LCNC technologies and CD practices in organizations. Thus, this study offers the following three key contributions:

- The systematic review provides a comprehensive analysis of the benefits and challenges of LCNC adoption and CD practices. The benefits of LCNC and CD include process improvement, profit maximization, digital skill development, empowerment, innovation development, usability, customizability, integration, maintenance, and scalability. On the other hand, the challenges of LCNC and CD practices relate to development platform fragmentation, third-party lock-in, low scalability, poor supportive architecture, unclearly defined development strategy, security risks, problems of IT governance, cultural complexity, development knowledge gap, role conflicts and collaboration, and performance evaluation problem. These issues arise because LCNC development platform vary across vendors, technical and user characteristics, and organisational DT requirements and maturity trajectories in determining problem-solution fit. However, our review indicates that there has been limited focus on studies investigating the performance evidence around the benefits of LCNC and CD adoption.
- The insights generated by this review also provides a framework of four key themes that serves as signposts for researchers and practitioners to consider when understanding LCNC and CD adoption practices. These themes focus on the nature of technology, management implementation strategies, employees' roles and digital skills, and performance evaluation metrics.
- Finally, our review helps identify research gaps and future opportunities associated with LCNC and CD adoption, providing researchers with a foundation to position their work for contributions in this domain.

The remainder of our article includes the following. Section 2 explains our research approach, while Section 3 presents our research results. Section 4 discusses the theoretical frameworks applied in the literature, and Section 5 highlights the methodological approaches applied in the literature. In Section 6, we offer an informative framework to guide the overall research contributions. In Section 7, we discuss our research agenda and then the conclusion and limitations of our study in Section 8.

2. Research method – A multi-phase systematic literature review

The section presents the SLR process that was adopted to identify and analyse all relevant research available for this study. The purpose of this SLR is to identify how the terms and labels based around LCNC and CD were used relevant to the business industry, and specifically to answer this study’s research questions (see Table 1). The SLR presented several key benefits to address research questions and identify the benefits, challenges, and opportunities of CD for DT.

As mentioned previously, we chose a grounded theory approach described by Wolfswinkel et al. (2013). The grounded theory consists of five stages; (i) define, (ii) search, (iii) select, (iv) analyse and (v) present. This provided an excellent approach to identifying relevant papers for this study in a manner that ensured clear, impartial, and repeatable reviews of large quantities of research papers (see Fig. 1). By performing an SLR using this approach, information related to the research questions of this study, and the different labels and definitions associated with CD and LCNC were identified and analysed. This research method ensured due diligence and would be a repeatable, unbiased method to be carried out in future studies.

2.1. Stage 1: Define

Define is the first of the five stages carried out for this SLR process and involved the initial search of literature that established the scope of this study. At this stage the specific (i) research fields to be included in the study were identified; (ii) the search engines and research sources to be used were decided upon; (iii) the search strings to be used were determined, and (iv) the inclusion and exclusion criteria to be followed were agreed upon and validated by the authors.

The fields of research identified as being appropriate for this SLR were in the domain of business, information technology, engineering, and information systems literature. It was essential to establish the appropriate fields of research as it was found that when searching the term ‘citizen development’ it resulted in several studies relating to ‘citizens’ in different fields. Being familiar with the research fields desired to be included in this study from previous research was very beneficial as studies resulting from unrelated fields were ruled out quickly. This stage ultimately resulted in the formulation of various possible search terms, which were further distinguished into search strings. Google Scholar was the search engine used for this initial search in order to identify some seminal or go-to papers (e.g., Waszkowski et al., 2019) with high citations as a starting point for the search process. For the stages following the initial search ACM Digital Library, AIS e-Digital Library, Science Direct, EBSCO (Academic Source Complete), Web of Science, Engineering Village and Scopus, as well as Google Scholar were the search engines and research sources used. The set of search terms in Table 2 is reflective of the whole scope of the research. It

Table 1
List of research questions.

RQ #	Research Question
RQ1	What are the benefits and challenges associated with the adoption of LCNC technologies and CD practices in organizations?
RQ2	How do organizations evaluate the methods they use in measuring the benefits of LCNC adoption and CD practices?

is worth noting that each search term (1 to 8) was applied in each database for generating our searched results.

Anticipating a high volume of studies from the next stage of this SLR process (search), inclusion and exclusion criteria were developed (see Table 3). The inclusion criteria were used to optimize the findings. The inclusion criteria and exclusion were applied to studies published over the past five years, i.e., between 2017 and 2023. As the concepts of CD and LCNC is relatively new, it was found from the initial search that there were not many studies related to this topic being published prior to those years. Conference papers were part of the inclusion criteria so that the most recent developments, theories, and trends in relation to this topic were included in the search. A log of the search criteria was kept with details of the resulting studies papers, to ensure that if revisions of the review process were needed there was sufficient documentation.

2.2. Stage 2: Search

This stage is when the actual search is carried out in the search engine databases using the defined search strings. We used our search strings in the selected search engines which yielded an overall sample of 5996 studies. The number of hits per search string and per database can be seen in Fig. 1. There were different volumes of results per database, which caused revisiting some of the search strings. For example, the terms software and technology were added to a search string so that all angles of the scope of our research would be covered. The large volumes of literature resulting from each search string used in each database were narrowed down to studies relevant to this SLR by reviewing the literature for inclusion on the defined search strings in the titles, abstracts, and keywords. All search engine databases used allowed for the use of ‘AND’ and ‘OR’ operators which were used for each search string as they contained more than one search term.

2.3. Stage 3: Select

This systematic literature review is based on empirical articles to provide a solid evidenced-based research in this area. Our analysis also identified key gaps in the literature and areas for future research. The reviewed literature was published between the years 2017–2023. The first search process took place between 22 February to 1 September 2022. However, an extended search period to 2023 took place between July and August 2024. We found 9 relevant additional empirical papers. While our focus was on searching for peer review papers in scientific databases only, the use of a grey database like Google scholar offered opportunities to retrieve and confirm relevant and highly cited seminar papers. Although many hits were generated running in thousands then, we narrowed them down by further “sorting by relevance” and using precisely the term low- code only or no-code only in the search text, for example. Overall, each database has their unique filters (e.g., journal sources on information and communication technologies, technological innovation, etc.) which we applied before we further applied our general filters (e.g., by title, abstract, etc.). At this stage, by removing 3912 duplicate and unsuitable papers from the searched 5996 papers, we were left with 2084 unique papers to be further reviewed. Firstly, the literature was filtered based on the inclusion of the defined search strings in the titles of the studies. This resulted in 454 papers included by reading the titles. These papers were saved into relevant folders for further review. We reviewed the abstracts of the 454 papers for the inclusion of the search strings. From this filtering process, 101 papers were included (353 excluded). Finally, the full texts of those 101 papers were further reviewed for relevance and 23 irrelevant papers were further removed based on the search strings, subject matter, discussions, etc. This resulted in the final selection of 78 relevant papers but only 40 of these were empirical studies (see Fig. 1).

The inclusion and exclusion criteria mentioned in Table 3 were used here for the selection of the final group of literature to be included in this SLR. We decided to exclude short papers with less than 5 pages without

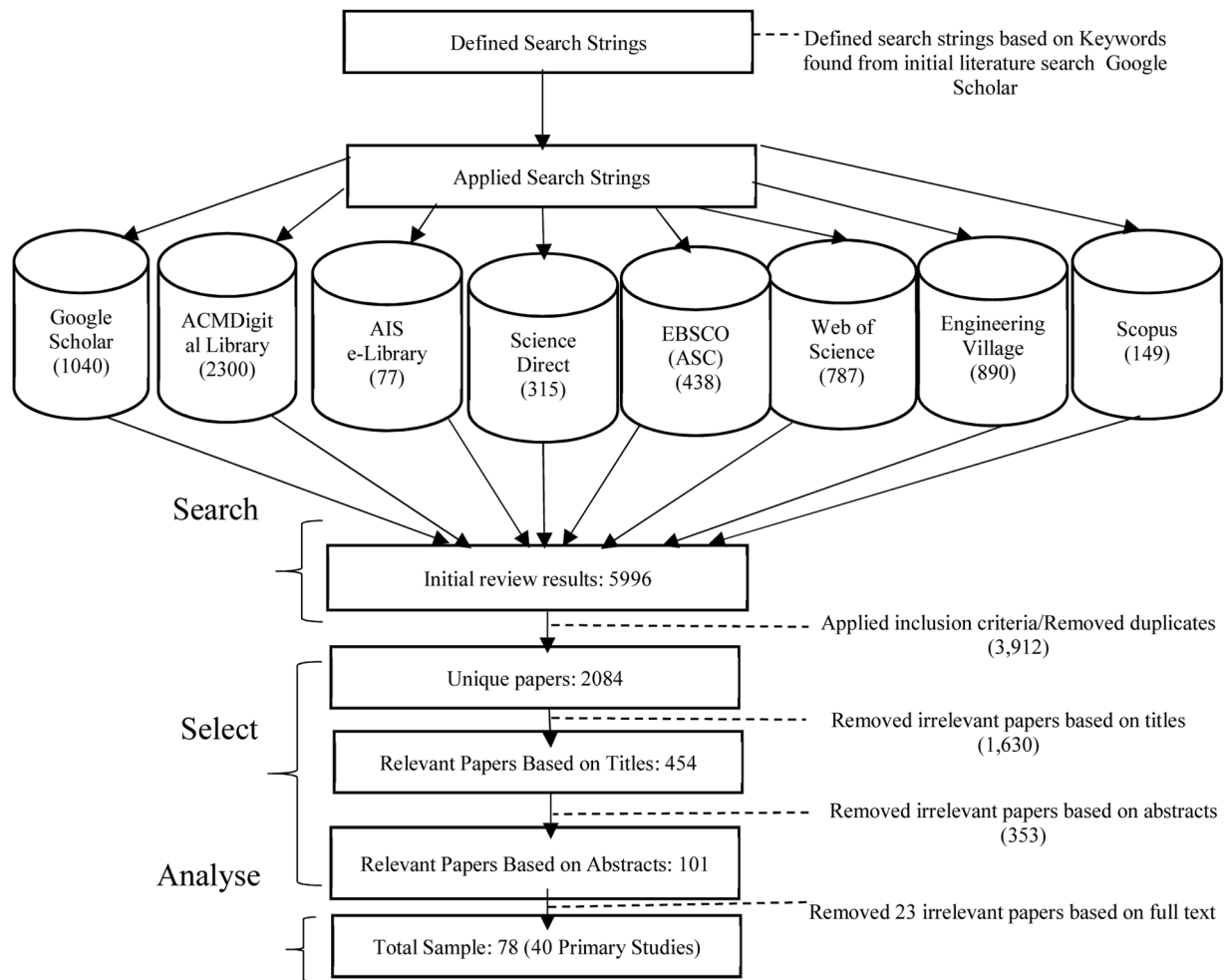


Fig. 1. Selection process of primary studies, incorporating stages 1–4 of the SLR approach.

Table 2
List of search terms.

Search Terms	
1	("citizen development" OR "low-code platform" OR "no-code platform" OR "low-code development platform" OR "no-code development" OR "low code-no code" OR "low code" OR "no-code" OR "low-code engineering" OR "low-code engineering platform" OR "LCNC" OR "citizen developer" OR "citizen oriented applications") AND ("digital transformation")
2	("citizen development" OR "citizen developer" OR "citizen oriented applications") AND ("digital transformation")
3	("citizen development" OR "citizen developer") AND ("digital transformation")
4	("low-code development platform" OR "low-code engineering platform" OR "low-code platform") AND ("digital transformation")
5	("no-code platform" OR "no-code development") AND ("digital transformation")
6	("low code-no code" OR "lcnc") AND ("digital transformation")
7	("citizen development" OR "low-code platform" OR "no-code platform" OR "low-code development platform" OR "no-code development" OR "low code-no code" OR "low code") AND ("digital transformation")
8	("no-code software applications" OR "low-code software applications" OR "no-code software" OR "low-code software" OR "no-code technology" OR "low-code technology") AND ("digital transformation")

methodological considerations to align with prior SLR studies (e.g., see Kitchenham et al., 2010, Zhou et al., 2016). Forward and backward citation tracking was checked on the selected studies to ensure they were the right fit and added to the quality of the studies gathered for this SLR. The reason for the inclusion of any studies outside of the criterion, from the forward and backward citation tracking, was discussed and agreed upon by all researchers.

Table 3
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
• The studies should be published between 2017 and 2023. • The studies should relate to any of the research questions of this study. • The studies should be published in the English language. • The studies have been published in scholarly, peer-reviewed journals, including conference papers.	• Studies not published in the English language. • Studies not related to the field of research of this study. • Non-scientific journals, e.g., keynote speeches, editorials, presentations, panel and editors' reports, experience reports, call for papers, research in progress, and lessons learned. • Duplicate articles and short papers. • Studies not available in full-text and PDFs for access and readership

2.4. Stage 4: Analyse

During this final stage of the SLR, the full texts of the selected papers were analysed. These papers were reviewed, with any relevant findings and insights into the scope of the SLR and the research questions, being noted, highlighted, and treated as excerpts for this study. Open, axial, and selective coding techniques were practised at this stage in the analysis of the data from the final selection of studies. In a similar method to that practised by Huber et al. (2022), this stage began with reading through the selected literature thoroughly and findings and identified excerpts relevant to this study were highlighted. Using axial coding themes relevant to the research questions posed in this study

were identified and then these themes were further synthesised through selective coding. Ultimately this led to the identification of benefits, challenges, and performance measures of LCNC reported by CD practitioners in the literature. These themes went through various iterations during this process until there was unanimity on the final themes, discussed in the following section, amongst the researchers. Further to this, the identified themes from the literature and the concepts and theories developed by the researchers involved in this study led to the creation of the framework discussed in the findings and analysis section of this study.

2.5. Stage 5: Present

The final stage of this SLR process has resulted in the presentation of this entire study. The findings and analysis in the following section communicate the results obtained from the SLR.

2.6. Threats to validity

Every SLR study is liable to threats to validity which could affect the results attained and ultimately the outcome of the study (Petersen et al., 2015). This section identifies the threats to the validity of this study and discusses the strategies used to alleviate the possibility of those threats.

2.6.1. Threat to construct and possibility of bias in literature selection

In this SLR we wanted to ensure that the search strings used would be sufficient to adequately measure the construct of this study. An adequate search string should retrieve as many papers as possible on the related topic. The results obtained from the use of the search strings prove the quality of the search string. To ensure validity, in relation to this threat, search strings were tested on our initial literature search and any revisions of the search strings to improve the recall of the results were documented (Edison et al., 2020). There is always the possibility of missing articles, but this is aimed to be avoided by revising the search strings and ensuring all keywords relevant to our search were included in different combinations (Wohlin, 2014). To ensure we did not miss any relevant papers we did not limit the results to specific domains related to this study. Instead, we allowed for the inclusion of all papers at that stage and then filtered them based on their relevance to the study as described in Section 2.3.

Furthermore, when researching literature in a specific field there can be a potential for bias from the researchers. This bias can result from the selection of papers based on their sources (i.e., the publishers used), the type of paper (i.e., a journal, a conference paper etc.) or if the results of the paper had positive or negative outcomes (Edison et al., 2020). To address this threat of bias the SLR process used for selecting papers did not limit the results based on publishers, or the types of resulting research papers. As long as the resulting papers were complete, peer-reviewed published articles that fit our inclusion criteria, were included. To further limit subjective bias, a review protocol was followed during this SLR, as described in the above sections by all researchers. This was to improve consistency in the selection of papers amongst the researchers and to ensure the inclusion and exclusion criteria were adhered to in the process.

2.6.2. Threat to internal and external validity

The results from a study that are produced from measured factors and from focusing on the research questions outlined in the study are referred to as internal validity. Threats to internal validity were combatted by only including results that came from the research questions set out by the researchers of this study. This ensures the results are controlled and measured and do not include results from factors outside of this study's research questions (Petersen et al., 2015). The generalization of an SLR, the findings and the overall outcome should be an objective for the researchers of any SLR as it attributes to the validity of the study, referred to as external validity (Petersen et al., 2015). To

combat threats to external validity the research methodology of this SLR followed a 5-step process, accurately described in the sub-sections above, that could be applied to studies in fields unrelated to this study.

3. General definitions of identified terms in the SLR

Findings relevant to the focused research questions of this study resulted from the analysis of 40 primary papers. The following definitions were identified from our SLR findings for this study.

3.1. Definitions of citizen development within LCNC platforms

Although our research indicates that there has been a growth in the adoption of CD and LCNC, one key finding from this study, which was mentioned in previous studies (Bock and Frank, 2021; Hoogsteen and Borgman, 2022; Luo et al., 2021) is the lack of clarity or agreement in defining LCNC and CD. The definitions, labels and abbreviations used for CD and LCNC (see Table 4) across the literature, as well as the references, varied greatly. It was also noted how LCNC was referred to as a tool, platform or set of tools (Beranic et al., 2020; Pichidienthum et al., 2021) in some studies. While in others it was discussed with a broader view and referred to as a 'low-code movement' (Cabot, 2020; Palomés et al., 2021; Pantelimon et al., 2019), as a 'paradigm' (Al Alamin et al., 2021), as an 'ecosystem' (Abidin et al., 2021), and as no-code development (Pantelimon et al., 2019). Our findings indicate that multiple definitions are highly ambiguous and therefore lack clarity conceptually regarding the concept of CD and LCNC development platforms.

Our findings also identified that low-code development platforms may be more suited for someone with minimal coding skills (Al Alamin et al., 2021; Indamutsa et al., 2021). Some studies have considered their suitability for someone with no coding skills (Bexiga et al., 2020; Lara-Prieto and Flores-Garza, 2021); whereas further studies have regarded both LCNC development platforms being suited to different individual needs (Abidin et al., 2021; Pantelimon et al., 2019). This lack of specificity, inconsistency, and ambiguity requires more conceptual clarity.

Overall, based on our SLR, we categorize our findings into four main themes (i) nature of technology development platforms, (ii) management implementation strategy, (iii) employees' roles and digital skills, and (iv) performance evaluation metrics. These categories offer sub-themes that guide our understanding of the benefits and challenges of CD in developing and deploying DT initiatives in organizations.

3.2. Definitions of efficiency within citizen development and LCNC platforms

Findings from our SLR revealed different forms of efficiency that were used to describe the benefits associated with the adoption of LCNC and CD solutions in organizations. CD solutions refer to solutions generated or derived from the utilization of LCNC technology that support DT initiatives, such as developing and deploying software applications to solve specific business problems or use cases in organizations. Therefore, most of our reviewed papers suggest that efficiency can be time-based, effort-based, collaboration-based, and financial-based. Time-based efficiency relates to wall-clock or run-time on a server to automate the deployment of a module or an application into production. This can also be described as computational efficiency (Korkan et al., 2020; Op't Land et al., 2021). Effort-based efficiency relates to the reduction in hours a developer takes to perform and complete a set of software development tasks such as writing specifications, coding, or programming for an application to be developed (Pacheco et al., 2021; Pantelimon et al., 2019). Collaboration-based efficiency relates to the ability of developers to start and complete a set of software development activities by working effectively together towards development goals (Pacheco et al., 2021). Lastly, financial-based efficiency here translates to a reduction in monetary and material costs of operations in software development such as labour costs (Bhattacharyya and Kumar, 2021;

Table 4
LCNC Definitions incorporating citizen development.

References	Definition Given
Abidin et al. (2021)	LCNC provides a diverse range of solutions and new challenges for anybody without code development experience or less formal training in programming.
Al Alamin et al. (2021)	Low-code software development (LCSD) is an emerging paradigm that combines minimal source code with interactive graphical interfaces to promote rapid application development.
Alsaadi et al. (2021)	The low-code development platforms help the developers to build fast, efficient, and scalable applications with a minimal need for coding, which introduced the concept of citizen developers in the field of application development.
Bexiga et al. (2020)	Low-code platforms [...] allow people with non-programming backgrounds to craft digital systems capable of solving business-relevant problems.
Da Cruz et al. (2021)	Low-code is an emerging concept that transforms visual representations into functional software, allowing anyone to be a developer.
Fill et al. (2021)	Low-code platforms can be regarded as the next step in the evolution of techniques for creating software applications with less effort for writing programming code.
Hoogsteen & Borgman (2022)	Low-code development platforms can help solve these issues [referring to project backlogs and shortage of developers] by using technology to empower end-users without a programming background to participate in the software development process. This trend is referred to as “citizen development.”
Hurlburt et al. (2021)	A low-code environment requires some programming acumen, while a no-code environment permits so-called “citizen developers” to assemble custom enterprise Software as a Service applications with little or no programming skills.
Iho et al. (2021)	By doing away with the necessity of writing code, low-code development platforms (LCDPs) offer to open software development to non-IT employees.
Lara-Prieto et al. (2021)	Low-code development platforms [...] enable people without a technological background to create and implement custom applications to improve specific processes.
Lourenço et al. (2021)	Low-code platforms aim at allowing non-experts to develop complex systems and knowledgeable developers to improve their productivity in orders of magnitude
Overeem et al. (2021)	LCDPs facilitate end-user programming for citizen developers, people without formal programming education that develop software, through Model-driven Development (MDD). (cited)
Palomés et al. (2021)	The low-code movement has recently emerged, aiming to remove barriers to using, configuring, and building software for everybody, enabling a general digital cultural literacy, and facilitating the challenge of interdisciplinary work
Pantelimon et al. (2019)	low-code / no-code application development [...] allowing the line-of-business professionals to build their own applications in the cloud without the hassle of writing code themselves
Silva et al. (2021)	They [referring to LCDPs] automate low-level and routine development tasks, effectively contributing to solving the problem of the global shortage of professional software developers.
Wang et al. (2021)	In [...] low-code development platforms, [...] the key logic is expressed in graphical models developed by “citizen developers” who have limited development skills.

Pichidienthum et al., 2021). While we have been able to identify and distinguish these forms of efficiency, they tend to be interwoven in generally describing organizational efficiency and productivity in software development process.

4. Themes associated with LCNC and CD

The results of our SLR are presented in four main themes, namely, the nature of technology development platforms, management implementation strategy, employee roles and digital skills, and performance

evaluation metrics.

4.1. Nature of technology development platforms

4.1.1. Benefits

4.1.1.1. Simple and usable design. One of the technological motivations for adopting LCNC development platforms is their simple and usable design configurations in automating business processes (Hoogsteen and Borgman, 2022). The literature highlights revealed that LCNC development platforms are built on graphical user interface models that allow users to ‘drag and drop’ building blocks or visual diagrams in developing and deploying business applications (Lebens et al., 2021; Wang and Wang, 2021). In doing so, LCNC reduces the complex nature of software design based on its interactive and intuitive visual application layer editors to synchronise data by hiding the design complexity of the actual coding and simplifying computational concepts behind the software development platforms (Abidin et al., 2021; Metrôlho et al., 2020). Hence, the simplification of the nature of software design makes it easy to instantly edit their functionalities and application layers in real-time for greater user experience (Iho et al., 2021; Lourenco et al., 2021; Schenkenfelder et al., 2023), and in exporting the development artefacts or components for future reuse in a cloud environment (Lourenco et al., 2021). In this line, LCNC development platforms provide accessible technological tools that make it easy and simple to use for citizen developers (business professionals) to master the technical know-how required to solve business problems without deep technical knowledge or coding skills, in rapidly developing business applications and transforming businesses in organizations (Iho et al., 2021; Reilly, 2021).

4.1.1.2. Customizable. The simple, convenient, and drag-and-drop visual application layers of LCNC development platforms provide opportunities for individuals and organizations to make better custom changes and prototyping for innovative business applications (Bhattacharyya and Kumar, 2021; Krejci et al., 2021; Schenkenfelder et al., 2023). Research has that customizing IT solutions to the needs of business enterprises can provide them with competitive advantages. Yet, arguably the cost of customization may be unaffordable for some businesses (Moskal, 2021). With user knowledge and skills of LCNC development platforms, insights from a process mapping study that involved interviews with 25 IT professionals found that LCNC reduced customization effort by 30% from 5 to 3 days in software development project planning activities (Bexiga et al., 2020). In addition, the literature highlights small and medium-sized enterprises are able to afford building customized applications “using templates and extensions” to address their business needs (Bexiga et al., 2020; Moskal, 2021; Reilly, 2021).

4.1.1.3. Secured communication. Citizen development is deemed to have supported a seamless embedding of access control policy and generation of code on (including the security of) IoT devices for building a secured enterprise service infrastructure (Korkan et al., 2020; Zolotas et al., 2018). Studies found that LCNC development technologies help create end-to-end supply chain management systems (Asawa et al., 2021), and streamline communication between vendors and suppliers during procurement processes (Bhattacharyya and Kumar, 2021). In addition, it fosters enhanced communication between business and IT professionals (Novales and Mancha, 2023). In the manufacturing context, the application server of the Aurea BPM low-code platform was built using various technologies (e.g., Google web toolkit, Oracle PL/SQL, etc.) to automate business processes. Other authors posit that LCNC technologies have dedicated APIs that make information exchange between components interoperable for communication between application interfaces (Michael and Wortmann, 2021).

4.1.1.4. Enable systems integration. Integration within a CD context

relates to the extent to which users can develop and configure low-code development platforms with external services (Al Alamin et al., 2021; Schenkenfelder et al., 2023), such as integrating external web registration processing, external API and email configurations (Al Alamin et al., 2021). For example, the literature suggests that a model-driven low-code development approach provides opportunities to integrate assistive services into the production toolchain to manage information processes where full automation cannot be achieved without human operators (Michael and Wortmann, 2021). Other authors found that low-code development platforms assisted in integrating IoT devices with IoT platforms (Pantelimon et al., 2019).

4.1.1.5. Maintainable. Considering the simplified and user-friendly design of LCNC development platforms, the literature highlights that business applications developed by both business and IT professionals are easier to maintain due to little or no lines of code required for maintenance (Bexiga et al., 2020). By using low-code development platforms, it is asserted that the ability to make instant changes due to changes in business requirements without writing complex codes on the already developed business applications makes them easier to maintain than traditionally developed software (Alsaadi et al., 2021). Thus, CD made maintaining the software application layer easy for organizations.

4.1.1.6. Scalable. Scalable here refers to the tendency of using different LCNC development platforms to support building large-scale applications for businesses in organizations. The literature suggests that the ability to use LCNC development platforms to drive solutions to daily business problems by automating workflows and integrating them with existing processes and different IoT devices, resolves development delays with traditional programming approach (Korkan et al., 2020), which provides some degree of scalability for large-scale process automation within and across departmental functions in organizations (Alsaadi et al., 2021). For example, by surveying a group of 161 students at a university, the use of OutSystems low-code platforms was deemed to have scaled the delivery of two software engineering modules for project-based learning in classroom settings by estimated 40 min with OutSystems overview compared with an estimated 24 hours' effort with Developing Web Apps. In addition, on a Likert scale of 5, the measurable impact increased from 3.5 to 4.4 on average per effort (Fernandes et al., 2020). Other studies have reported the scalability potential of various LCNC development platforms in different settings in organizations (Metrolho et al., 2019; Zolotas et al., 2018). Despite these benefits, it is important to present a balanced view and outline its associated challenges reported in the SLR, as follows.

4.1.2. Challenges

4.1.2.7. Fragmentation of development platforms. LCNC development platforms are software technologies. There are many LCNC development platforms with different features such as Kissflow, TrackVia, Betty Blocks, Microsoft PowerApps, OutSystems, Mendix, etc. (Iho et al., 2021; Kholiya et al., 2021) that many organizations use to automate their workflow and business processes. Existing literature highlights the fragmented nature of these software tools. This is due to inconsistent development requirements, as well as claims around the derived benefits of some platforms and features over others, in supporting the development of business applications such as IoT systems (Alsaadi et al., 2021; Korkan et al., 2020; Oteyo et al., 2021). Considering that most development platforms are owned by third-party vendors, studies have argued in support of delivering a unified integration between different development tools, aimed at driving a better user experience with usable, friendly, and secured business applications (Alsaadi et al., 2021; Hoogsteen and Borgman, 2022).

4.1.2.8. Third-party lock-in. As many organizations often depend on

third-party vendors for LCNC development platforms to develop their business applications, the literature reports that there is a fear and risk of being locked-in by platform providers (Prinz et al., 2022; Schenkenfelder et al., 2023). The rising demand for LCNC development platforms for building business applications also risked huge costs of investment and the difficulty to switch platform vendors. This is also exacerbated when platform vendors discontinued supporting such platforms for continued operations (Alsaadi et al., 2021; Cabot, 2020; Pantelimon et al., 2019).

4.1.2.9. Low scalability. While studies have considered LCNC development platforms to be scalable in building business applications (Ali et al., 2021; Alsaadi et al., 2021), studies suggest that using such platforms would require advanced knowledge to enable large-scale computations and cloud resources which can be very expensive to achieve for development (Ali et al., 2021; Marek et al., 2021).

4.1.2.10. Poor supportive architecture. Studies suggest that LCNC development architectures can be limited and poorly constructed, which may resulted in the development of unusable applications that may deserve greater demand for employing more specialists to manage maintenance and transferability problems for citizen developers (Hoogsteen and Borgman, 2022; Novales and Mancha, 2023).

4.2. Management implementation strategy

4.2.1. Benefits

4.2.1.11. Process improvement. Process improvement here can be described from the perspective of business automation. Business automation involves the automatic processing and interaction of data entered into digital systems from different integrated sources by using digital technologies to perform and facilitate the smooth flow of business operations or workflow activities (Wang and Wang, 2021). A workflow is triggered when for instance an individual files a loan application online, which will in turn activate an immediate response by sharing information, for example, with the case manager until the loan application is approved electronically (Wang and Wang, 2021). Hence, business automation is about automating to improve and optimize business (workflow) processes through the use of low-code development platforms for the DT initiatives (Beranic et al., 2020; Lebens et al., 2021).

In sum, the literature highlights the use of emerging technologies such as LCNC development platforms for automating business processes instead of manual operations, thereby making CD capable of transforming workflows into more efficient business operations and services for users in various sectors (Elshan et al., 2023; Lebens et al., 2021; Wang and Wang, 2021). For example, studies suggest that by using low-code development platforms, non-technical farmers can develop technical skills to quickly automate their farming processes, such as building smart agriculture applications with temperature and humidity sensors to efficiently collect and process information about soil moisture contents. This would be to inform farmers' decisions and actions in planting, watering, tillering, and harvesting seasons (Oteyo et al., 2021).

Moreover, studies suggest that the realization of business automation is built on intelligent process automation (IPA). IPA's functionality is configured on robotic process automation (RPA) and artificial intelligence (AI) technologies. These technologies makes it possible for organizations to process data and automate complex processes beyond work routines in a more "quicker, reliable, and efficient" way than humans can perform (Kholiya et al., 2021). It is this robotic process automation and meta-modelling techniques that makes it possible to operationalize or transform manual tasks into digital forms (Bexiga et al., 2020; Kedziora et al., 2023; Lebens et al., 2021). Conclusively, with LCNC technologies for business automation, organizations can make smart

decisions to deliver operational efficiency and profitable businesses in the DT journey. In addition, process improvements are associated with time, effort, finance (monetary cost), and collaboration efficiencies.

4.2.2. Challenges

4.2.2.12. Unclearly defined development strategy. Our research findings indicate that poor strategy prevents the adoption of CD (Bhattacharyya and Kumar, 2021). The literature highlights suggest that a lack of cultural solidarity and poor organizational-wide awareness (Bhattacharyya and Kumar, 2021) and high learning curve (Schenkenfelder et al., 2023) limit the chances of LCNC and CD success for DT initiatives. Some studies suggest that organizations need to learn and understand the benefits, requirements, and challenges of employing LCNC digital solutions and how they support their DT strategy before adopting them for business transformation (Novales and Mancha, 2023; Hoogsteen and Borgman, 2022; Metrólho et al., 2020; Wang and Wang, 2021). Overall, the core argument is that merely adopting LCNC and CD practices is doomed to be ineffective or fail without clear purpose in understanding the problem-solution fit, organization-wide awareness for management and employee support, and a set of proper guidelines for checks and balances surrounding its adoption.

4.2.2.13. Security risks. Given the benefits of adopting LCNC development solutions to solve workflow problems and improve operational efficiency by easily and rapidly developing maintainable business software applications, researchers and experts warn of the risks it poses to organizations (Al Alamin et al., 2021). These concerns include risks of data security breaches and cyberattacks that can occur due to shadow IT risks where a citizen developer may develop a functioning business application without being sanctioned (a non-compliance issue) by the IT department (Elshan et al., 2023; Hoogsteen and Borgman, 2022; Oltrogge et al., 2018). As most organizations often depend on LCNC development platforms that are developed and controlled by third-party vendors; it leaves non-IT sanctioned use of LCNC digital solutions problematic for data privacy, quality control, maintenance, and governance flaws (Elshan et al., 2023; Hoogsteen and Borgman, 2022; Novales and Mancha, 2023; Oltrogge et al., 2018). For example, in managing online transactions, where there are little or no options to make custom changes, users may need to “fully trust the service to generate non-vulnerable code and do not include hidden or non-obvious user tracking or data leakage” (Oltrogge et al., 2018, p. 646). Hence, it is important to carefully consider the benefits and perceptions of the potential risks involved in adopting CD practices (Hoogsteen and Borgman, 2022). Indeed, Elshan et al. (2023) argues that while speed and fewer resources are beneficial for LCNC and CD practices in accelerating the development of applications, prioritizing speed may risks adhering to quality requirements that may be needed for superior product performance. Hence, this can result to technical debits in long term technology investments. Interestingly, the freedom for citizen developers to develop and deploy business applications without being conscious of cyber threats and network security, and the inability to detect where the boundary is, in cloud environment for software production, can defeat CD purpose if not well governed by IT department in organizations.

4.2.2.14. Problems of IT governance. IT Governance can be related to making and enforcing decisions to promote a safe, accountable, and acceptable use of technologies for business solutions in organizations. Some researchers and practitioners have considered the problem of governance as a critical challenge in impeding CD adoption and good collaboration among citizen developers and traditional software developers in organizations (Hoogsteen and Borgman, 2022; Iho et al., 2021; Prinz et al., 2022). Overall, the literature highlights that a lack of rules, standards, guidelines or directives for citizen developers can lead to inefficiencies and other system integration (Hoogsteen and Borgman,

2022) and knowledge integration problems (Iho et al., 2021). Other concerns raised include irrelevant software applications, security, and maintenance problems (Hoogsteen and Borgman, 2022). These studies suggest that a harmonising IT strategy and governance framework will be needed to address non-compliance issues in order to efficiently and effectively manage CD adoption and cross-functional collaboration of developers in the DT journey (Hoogsteen and Borgman, 2022; Iho et al., 2021; Novales and Mancha, 2023; Prinz et al., 2022; Stamoulis, 2022).

4.2.2.15. Cultural complexity. Organizational culture relates to how employees behave based on a set of norms and values dear to the organization. In the era of CD, the literature highlights that the complexity of culture (e.g., to raise awareness/communicate development knowledge) can affect work relations between industrial domain experts and professional software engineers, which can subsequently affect the success rate of software development (Wang et al., 2021). It is however noted that the longstanding cultural barrier between business and IT units may be due to the different ways by which business and IT professionals work in organizations (Iho et al., 2021). Others have even suggested a technological culture (Thacker et al., 2020) and data-driven culture (Stamoulis, 2022) to promote data analytics and CD adoption in using LCNC development platforms for driving DT initiatives.

4.3. Employee roles and digital skills

4.3.1. Benefits

4.3.1.16. Employees' digital skills development. Employees' digital skills development relates to reskilling and upskilling of individuals with the appropriate information, communication, and technology (ICT) skillsets required for the DT process in IT and software-oriented organizations. The reason for this is that many organizations are challenged with a shortage of software developers or IT professional skills (Hurlburt, 2021; Metrólho et al., 2020; Novales and Mancha, 2023; Op 't Land et al., 2021). Hence, the need to acquire and pay for expert development skills in creating and maintaining user-driven applications for automating manual processes becomes essential (Lebens et al., 2021; Metrólho et al., 2019; Palomés et al., 2021). The literature highlights some technical and soft skills related to transforming businesses in organizations and the role of CD in using LCNC development technologies in mitigating some of these challenges (see Table 6). Among various digital-related specific skills outlined include computational thinking (Hylton et al., 2021; Palomés et al., 2021; Wang and Wang, 2021), creativity/innovativeness (Lara-Prieto and Flores-Garza, 2021; Wang and Wang, 2021) and understanding user experience (Metrólho et al., 2020) towards the use of software applications for business development in digital transformation initiatives (Metrólho et al., 2019; Wang and Wang, 2021). The adoption of LCNC and CD practices for DT process raises the level of cultural awareness for digital literacy among business professionals to develop some of these skills, which they might have not been exposed to originally. This can help researchers and practitioners to separate traditional and contemporary skills and approaches for software development activities.

Table 5 lists the common technical and soft skills reported in the SLR to deliver technology solutions within businesses. This also acknowledges and emphasizes the necessity for employees to develop the technical ability to use technologies or software-related tools in developing software products. These include creating and making sense of data, writing codes, and using a low-code interface development environment for developing software applications for business solutions. These specific skills offer overarching dimensions to a set of technical skills about software development requirements right from ideation to designing and developing product ideas into workable software solutions.

The soft skills outlined above in Table 5 reveal some important people or interpersonal skills that are not directly tied to the use of

Table 5
Digital-related specific skills.

Skills	Examples	References
Technical skills	Implementation, development, and data mastery of information systems/technology solutions	Bernsteiner et al. (2022), Metrolho et al. (2019), Schneid, Stapper, Thone, & Kuchen (2021)
	Design thinking & knowledge in creating intuitive applications	Bernsteiner et al. (2022), Lara-Prieto & Flores-Garza (2021), Metrolho et al. (2019), Thacker et al. (2020)
	Use of sensors and low-code interface	Colantoni et al. (2020)
	Coding/programming	Bernsteiner et al. (2022), Fill et al. (2021), Lebens et al. (2021), Lourenco et al. (2021), Luo et al. (2021), Hylton et al. (2021), Marek et al. (2021), Moskal (2021), Oteyo et al. (2021), Pantelimon et al. (2019), Schneid et al. (2021)
	Business app development	Metrolho et al. (2019), Prinz et al. (2022), Wang & Wang (2021)
	Operational technologies (OT), artificial intelligence, machine learning, computer vision	Lara-Prieto & Flores-Garza (2021)
	Computational thinking related to concept definition, system, and problem analysis (identifying/evaluating/solving problem)	Hylton et al. (2021), Marek et al. (2021), Metrolho et al. (2019), Palomés et al. (2021), Wang & Wang (2021)
	Computer science concepts/conceptualization	Colantoni et al. (2020), Hylton et al. (2021), Palomés et al. (2021), Thacker et al. (2020), Silva, Vieira, Campos, Couto, & Ribeiro (2021), Wang et al. (2021)
	Data analytics, theoretical knowledge of practical frameworks/solutions	Lara-Prieto & Flores-Garza (2021), Metrolho et al. (2019)
	Understanding user experience	Bernsteiner et al. (2022), Metrolho et al. (2020)
Soft skills	Critical thinking/planning technology solutions	Elbatany & Succi (2021), Metrolho et al. (2019)
	Self-regulation (self-monitoring)	Wang & Wang (2021)
	Creativity/ innovativeness	Bernsteiner et al. (2022), Lara-Prieto & Flores-Garza (2021), Stamoulis (2022), Wang & Wang (2021)
	Communication, presentation	Bernsteiner et al. (2022), Lara-Prieto & Flores-Garza (2021), Satuluri & Radhika (2021)
	Comprehension, time management, logical reasoning	Bernsteiner et al. (2022), Metrolho et al. (2019)
	Understanding teams	Bernsteiner et al. (2022)

technologies and software tools but can help to innovate and develop digital applications to fuel a DT process. These skills are often needed to support how various stakeholders and software developers comprehend and communicate problems, as well as to create business ideas and innovative designs in successfully managing software development projects. In essence, this research indicates that technical and soft skills are required in developing and deploying digital solutions to business processes for DT.

Many educational institutions of higher learning are making growing efforts to educate future graduates with technical skills (e.g., use of LCNC software technologies) required to develop business applications using Figma (Abidin et al., 2021), Microsoft PowerApps, (Baumgarten et al., 2020; McHugh et al., 2023; Wang and Wang, 2021) and Out-Systems development platforms (Fernandes et al., 2020; Metrolho et al., 2019; Metrolho et al., 2020), among others, to become more employable in the job market. This is also expected to fill the digital skills gap in

many industries, which include logistical supply chain (Asawa et al., 2021; Bhattacharyya and Kumar, 2021) and agricultural sectors (Oteyo et al., 2021) undergoing DT. For example, the literature reports that XXI Portuguese Government is aggressively pursuing the National Digital Competence Initiative e.2030 through the Science Technology Engineering and Mathematics (STEM) programme. It is anticipated that this will see Portuguese citizens becoming digitally skilled and employable in using digital technologies and applications for the socio-economic growth and competitiveness of Portugal in Europe (Metrolho et al., 2019).

Overall, CD is considered important to address the shortage of traditional software developers by providing business professionals with training opportunities and the ability to learn, develop, and employ new technical and soft skills with little or no coding experience in driving digital solutions to business problems (Metrolho et al., 2019).

4.3.1.17. Empowering employees. Empowering employees can be described as providing autonomy to employees at work in organizations. Employee autonomy reduces the dependency on others with more latitude in decision-making processes that affects work performance. In this instance, CD has been found to empower business employees and support organizations not to over-depend on traditional software developers (i.e., programmers or coders) using IT-sanctioned LCNC technologies to solve their everyday business problems. This is often done by championing business applications and bridging the gap between business and IT units (Bhattacharyya and Kumar, 2021; E. Elshan et al., 2023). The literature highlights found that empowered employees become more creative, contribute better to knowledge sharing and work in a hyper-agile way to develop ideas and drive technological solutions for digital innovation (Abidin et al., 2021; Carroll and Maher, 2023; Hoogsteen and Borgman, 2022; Krejci et al., 2021; Novales and Mancha, 2023). This unconventional way of working by citizen developers and organizations thus provides some degree of autonomy, flexibility, and agility to make quick but good decisions in adapting to the rapid changes in environmental demands and requirements for software implementation. However, this also reduces the influence traditional software developers have in the software development process who often code from scratch to produce software products (Bhattacharyya and Kumar, 2021; Bock and Frank, 2021; Op 't Land et al., 2021). Table 6 provides the benefits of CD to employees in organizations.

Overall, CD is focused on empowering employees outside IT departments to develop ideas on IT-sanctioned LCNC development platforms (Bhattacharyya and Kumar, 2021; Krejci et al., 2021). However, in the absence of support tools on LCNC development platforms to develop certain business requirements, employing the programming skills of software developers becomes ever important.

4.3.1.18. Knowledge integration. Knowledge integration relates to “applying and synthesizing specialised knowledge residing in disparate parts

Table 6
Empowerment benefits of citizen development.

Empowerment Benefits	References
Business agility/flexibility/adaptability	Bhattacharyya & Kumar (2021), Bock & Frank (2021), Carroll & Maher (2023), Elshan et al. (2023), Novales & Mancha (2023), Op 't Land, Krouwel, & Gort (2021), Hoogsteen & Borgman (2022), Krejci et al. (2021), Varajão et al. (2021)
Work freedom/autonomy	Krejci et al. (2021)
Smart decision-making process	Hoogsteen & Borgman (2022)
Intellectual/creative potential	Hoogsteen & Borgman (2022)
Hyper-agile idea development process/actualization	Abidin et al. (2021), Carroll & Maher (2023), Krejci et al. (2021)
Higher employability, organizational resilience, minimize risks, quick problem-solving practices	Bhattacharyya & Kumar (2021), Hoogsteen & Borgman (2022), Krejci et al. (2021)

of an organization" (Iho et al., 2021, p. 1). The gap between business and IT units has been a long-standing problem and has made knowledge integration challenging for business and IT professionals (Iho et al., 2021). This is because to integrate knowledge, business and IT professionals need to share knowledge to solve problems and increase productivity (Lebens et al., 2021; Lourenco et al., 2021; Metrolho et al., 2019). The literature highlights that CD closed this knowledge transfer gap by enabling business professionals to combine their business knowledge with LCNC technology capability in solving problems and developing innovative solutions for DT (Iho et al., 2021; Pantelimon et al., 2019). For example, Iho et al. (2021) interviewed IT professionals and reviewed 201 user reviews from 12 different low-code development platforms, they found that low-code development platforms helped them to integrate their business knowledge with technical functionality of LCNC platforms playing a pivotal role in contributing towards business innovations in their organizations (Hoogsteen and Borgman, 2022).

4.3.1.19. Idea development. Organizations are often time-pressed to innovate to meet the changing needs of customers as well as compete with rivals in their business environment. In this vein, employees are saddled with the task of innovating at high speed to create and sustain business value for the organizations where they operate (Iho et al., 2021; Stamoulis, 2022). This in essence compelled business domain professionals to network for IT solutions with IT professionals before turning their ideas into workable prototypes. CD assisted business professionals to reduce these barriers by instantaneously developing their software ideas into prototypes for potential new business innovations by neither needing a connector nor leaving it to the rest of the IT professionals to achieve such realization (Fernandes et al., 2020; Krejci et al., 2021; Lebens et al., 2021; Palomés et al., 2021). For example, within the application of low-code engineering techniques, observations and interviews with 7 industrial domain experts revealed that they are happy to develop their industrial software applications without consulting with their professional developers (Wang et al., 2021). According to the work of Krejci et al. (2021), in a study of 10 expert interviews with executives and employees (e.g., senior developer, platform developers), and analysed reports from analysts working in 12 different low-code development platforms, it was found that LCNC empowers business professionals with greater capability to generate and develop ideas for digital innovation.

4.3.2. Challenges

4.3.2.20. Development knowledge gap. Knowledge (in terms of technical and soft skills) is a great resource in performing software development activities such as addressing design problems and using LCNC development tools to build software (business) applications (Thacker et al., 2020). In general, business knowledge and technical knowledge have remained dominant in fulfilling software development requirements (Iho et al., 2021). Despite the trending rise in demand for LCNC development solutions for improving business processes and workflows; some studies report that there is a skills gap in the knowledge application of LCNC development platforms (Hoogsteen and Borgman, 2022; Krejci et al., 2021; Silva et al., 2021). These studies suggest that a lack of appropriate software-related skills may be a drawback for software and IT-oriented organizations to address data, user experience, automation and downtime development issues toward DT (Op't Land et al., 2021). Table 5 offers insights into a list of digital and software-related knowledge skills (e.g., design/computational skills), such as understanding how to use low-code interfaces for business app development (Abidin et al., 2021; Lourenco et al., 2021; Marek et al., 2021; Wang et al., 2021).

4.3.2.21. Role conflicts and collaboration. Conflicts of interest, loss of power or influence, and dissatisfaction are bound to create tension in the

way of working among professionals. In the case of CD, where business professionals are empowered to use LCNC development technologies with little or no intervention by IT professionals, the literature highlights that this can create conflicts as IT professionals lose control or have reduced influence in driving business professionals' work performance (Al Alamin et al., 2021; Lebens et al., 2021). This can further affect the degree of collaboration between them on development tasks, particularly where there is no mutual agreement or consensus over data security issues due to business applications developed by citizen developers without being supported or sanctioned by the IT professionals either for a legitimate reason or not. This can create resentment toward fostering sustainable collaboration between business and IT professionals or business and IT units. For example, studies revealed that traditional software developers were no longer able to satisfy recent industrial software development needs due to shifting work patterns as citizen developers gain more control and ownership of their software development activities in an organization (Prinz et al., 2022; Wang et al., 2021). Despite citizen developers' lack of coding skills, the use of a low-code development environment was found to help them in specifying task workflows, accessing the repository, selecting services to automate, configuring triggers and actions, and finally authorising task workflows (Indamutsa et al., 2021). This tends to reduce the dominance and the impact of traditional software developers, while the citizen developers become more empowered to take a leading role in the software development process.

4.4. Theme 4: Evaluating the performance benefits of LCNC and CD solutions

4.4.1. Benefits

4.4.1.22. Time-based efficiency. In light of the literature highlights, and given that LCNC development platforms required little or no coding requirements; CD has been found to shorten the length of time needed to develop and deploy software applications (Alsaadi et al., 2021; Bexiga et al., 2020; Bhattacharyya and Kumar, 2021; Pacheco et al., 2021). One study in particular examined 25 LCNC users and 3 developers in an observatory and laboratory experiment and found that low-code development platform reduced the time it takes to create a module generator by 20%. In doing so, they were able to generate automatic codes that support the development and deployment of enterprise software modules (Pichidienthum et al., 2021). Likewise, a survey of 20 industrialists were found to have reduced the development time from 15 days to less than 1 day in generating and deploying smart contracts on the Ethereum blockchain as solutions to contract problems in supply chain management. Although it was not clear whether some of these industrialists were also direct users of LCNC platforms (Asawa et al., 2021).

4.4.1.23. Effort-based efficiency. The major comparison between the use of traditional software development and LCNC software development platforms is that the former requires a great deal of manual effort to write codes for developing, testing, and deploying complex software products. The latter requires automatic generation of code, and where manual coding is required for customizations, it is at a low level based on the feature of a drag-and-drop graphical user interface, otherwise, refer to as what-you-see-is-what-you-get (WYSIWYG) development platforms (Abidin et al., 2021; Alsaadi et al., 2021; Bexiga et al., 2020; Oteyo et al., 2021). This differentiates the work rate expectations for traditional software developers and citizen developers.

In one of the reviewed studies by Bexiga et al. (2020), from interviewing 25 IT professionals, it was found that the use of OutSystems' low-code development technology saved the development teams between 20 and 75% in the effort applied to designing a complex UX/UI design during the development process, in terms of deploying and

managing database scripts and application updates in a cloud environment (Fernandes et al., 2020; Metr  lho et al., 2020). In sum, CD is considered to have played a good role in reducing the human effort required to code, model, develop, and manage software applications. The efforts saved through LCNC platforms supported traditional software developers to attend to other crucial and more complex tasks such as installing, configuring, training and governing software systems and applications (Alsaadi et al., 2021).

4.4.1.24. Financial-based efficiency. Software is often very expensive to produce by traditional development methods. This is because writing complex codes and computations require the assembling of software development teams (Ali et al., 2021; Marek et al., 2021; Moskal, 2021). Hence, the time, effort and other resources required to develop and manage software applications are financially demanding (Asawa et al., 2021; Schneid et al., 2021). The literature indicates that CD reduced the budget or monetary cost of hiring a team of software developers in developing and maintaining software applications (Alsaadi et al., 2021). The literature findings also suggest that this can also lead to a reduction in software development time when utilising LCNC development platforms by lowering the budget or costs of producing software applications (Asawa et al., 2021; Hoogsteen and Borgman, 2022; Luo et al., 2021).

4.4.1.25. Collaboration-based efficiency. Collaboration explains how individuals communicate and coordinate work together to achieve performance (Pacheco et al., 2021). However, the literature highlights revealed that inefficient collaboration exists between industrial domain experts and professional software engineers (Wang et al., 2021); and between front-end developers, user experience (UX) and user interface (UI) designers (Bexiga et al., 2020; Pacheco et al., 2021). One common problem mentioned by these authors is that traditional software development techniques make it less effective to achieve an efficient collaboration process among professional software development teams (Bexiga et al., 2020; Pacheco et al., 2021; Wang et al., 2021). For example, it was noted that the transformational traditional design process of developing customer-facing applications is often done 100% manually by sketching, drawing, and transforming images. Unfortunately, the literature reported that this is prone to errors despite the lengthy time and effort required for sharing ideas and recruiting more collaborators for image modelling. In contrast, using the OutSystems low-code development platforms, it was found that a similar effort increased the number of application pages and screens deployed by 150% to 400%. Thus allowing more efficient and productive collaborative effort among software development teams (Pacheco et al., 2021). CD, therefore, improves the ability of both domain experts and enterprise users to smoothly collaborate during the software development process (Carroll and Maher, 2023).

4.4.2. Challenges

4.4.2.26. Problem of evaluating performance benefits of CD within LCNC platforms. Despite the numerous benefits of LCNC and CD practices in transforming business processes to achieve operational performance, only 9 out of 40 primary studies have provided some information on measurements for evaluation of its beneficial impacts on businesses and organizations (see Table 8 in Appendix). Table 7 and Fig. 2 presents a list of the challenges of CD and LCNC adoption.

In review, Table 8 indicates the state-of-research surrounding the evidence of the benefits of LCNC and CD adoption. This table offers an overview of the strengths and weaknesses of each study, potential gaps for research opportunities, and even in making contributions to practice. To reduce information overload in this table, we present the data visualization of our systematic review findings as depicted in Fig. 2, 3, 4, 5 Fig. 6. More importantly, in terms of the evidence backing LCNC and CD

Table 7
Summary of challenges associated with LCNC and CD adoption.

Fragmentation of development platforms	Inconsistent development requirements	Alsaadi et al. (2021), Korkan et al. (2020); Oteyo et al. (2021)
Third-party lock-in	Dependence on platform vendors for software updates/security	Carroll & Maher (2023), E. Elshan et al. (2023); Novales & Mancha (2023); Prinze et al. (2022); Schenkenfelder et al. (2023)
	High costs of investments	Alsaadi et al. (2021), Cabot (2020); Pantelimon et al. (2019)
Low scalability	Limited options for large-scale computations/cloud resources	Ali et al., 2021, Marek et al., 2021
Poor supportive architecture	Maintenance/transferability problems Single-based modelling system/components	Novales & Mancha (2023); Hoogsteen & Borgman (2022) Hoogsteen & Borgman (2022)
Unclearly defined development strategy	Lack of cultural solidarity	Carroll & Maher (2023); Bhattacharyya & Kumar (2021)
	Limited organisation-wide awareness Low understanding of benefits/ challenges	Bhattacharyya & Kumar (2021) Hoogsteen & Borgman (2022), Metrolho et al. (2020), Wang & Wang (2021)
Security risks	Data security breaches/ cyberattacks, Shadow IT/non-compliance issues	Elshan et al. (2023), Hoogsteen & Borgman (2022); Novales & Mancha (2023); Oltrogge et al. (2018)
Problems of IT Governance	Lack of rules/standards/ guidelines/ directives	Carroll & Maher (2023), Hoogsteen & Borgman (2022); Novales and Mancha, 2023 Hoogsteen & Borgman (2022)
	Knowledge integration/ maintenance problems Difficult pathway to communicate development knowledge	Carroll & Maher (2023), Wang et al. (2021)
Development knowledge gap	Lack of appropriate software skills	Op' t Land et al. (2021)
Role conflicts and collaboration	Loss of control/reduced influence of IT professionals on business professionals	Al Alamin et al. (2021), Lebens et al. (2021)
Performance evaluation problem	Limited and inconsistent performance metrics	Ali et al. (2021), Asawa et al. 2021; Bexiga et al. (2020); Fernandes et al. (2020); Pacheco et al. (2021); Pantelimo et al. (2019), Pichidtionthum et al. (2021); Silva et al. (2021)

benefits, it is clear that there is a disjointed view of how organizations have measured the performance benefits (e.g., time-based efficiency, effort-based efficiency, etc.) of CD solutions within LCNC platforms. It is also safe to say that there are inherent overlaps that maybe be inevitable by which effort-, time-, and financial-based efficiency can be interpreted. As LCNC and CD practices represent a new phenomenon, it may not be surprising to find rare evidence of measurements and the lack of generalised metrics for evaluation. Our SLR shows that out of 40 empirical studies, only 9 studies had evaluation criteria. Thus revealing the scanty empirical evidence surrounding the measurements of CD benefits, as well as some useful information about the way organizations have measured their performance.

5. Applied theories and methodological approaches in empirical studies

Citizen development is a new research phenomenon that is gaining traction in academic and practitioner communities. While practitioners typically focus on the model configurations, applications, and

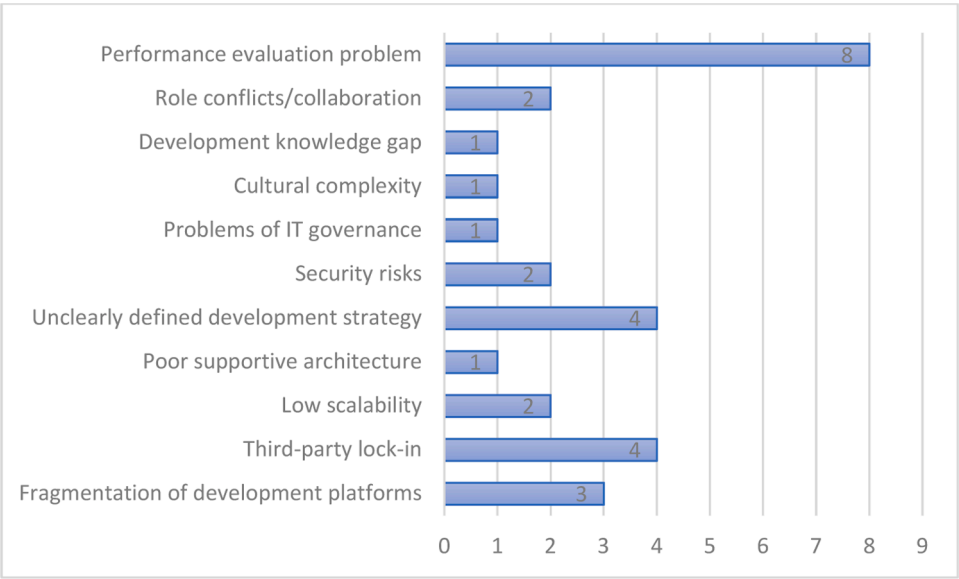


Fig. 2. Challenges.

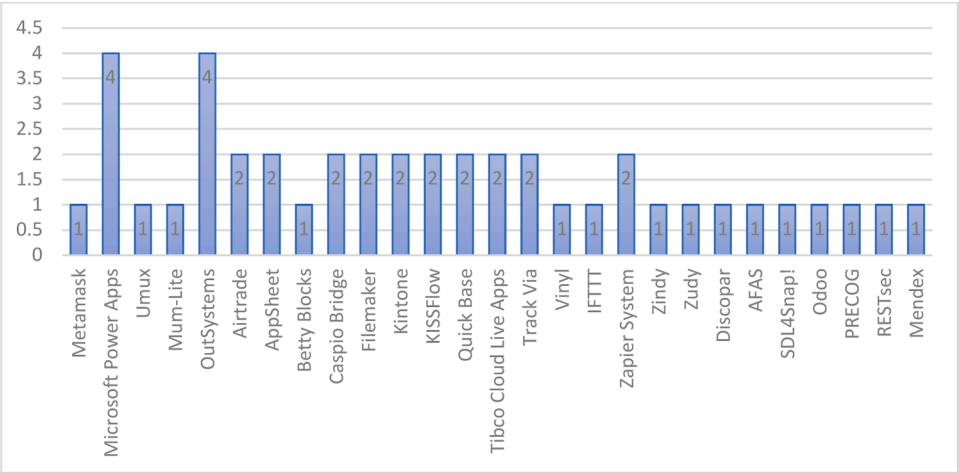


Fig. 3. LCNC platforms.

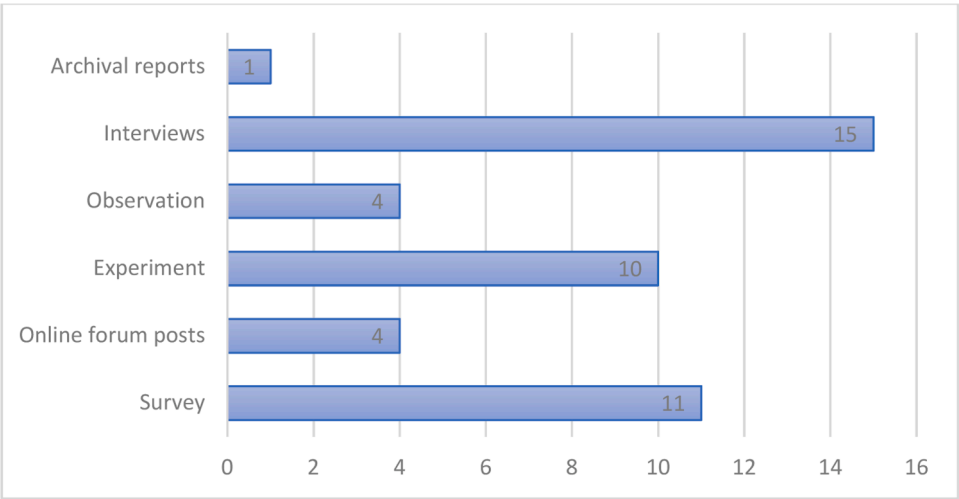


Fig. 4. Evaluation methods.

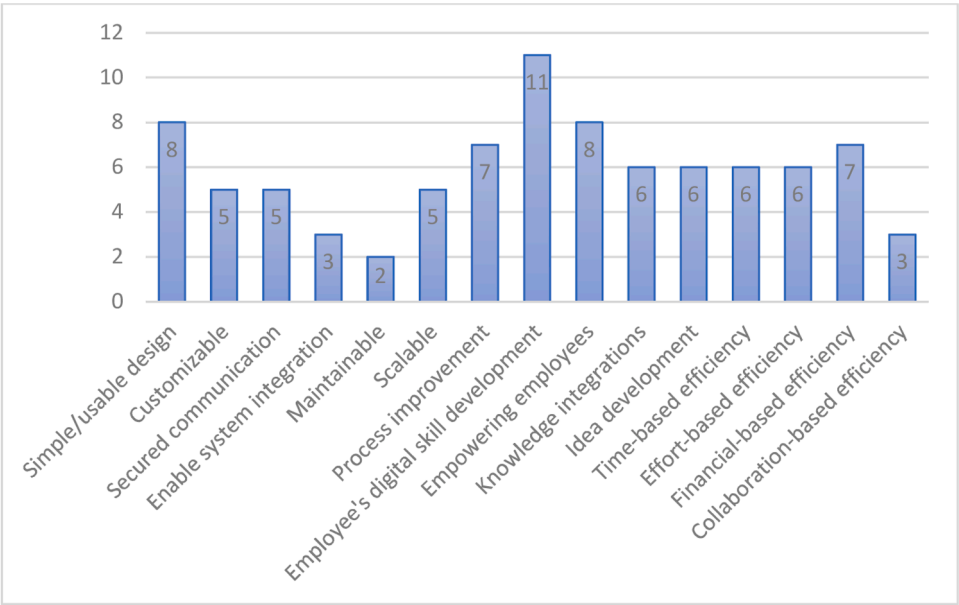


Fig. 5. Benefits.

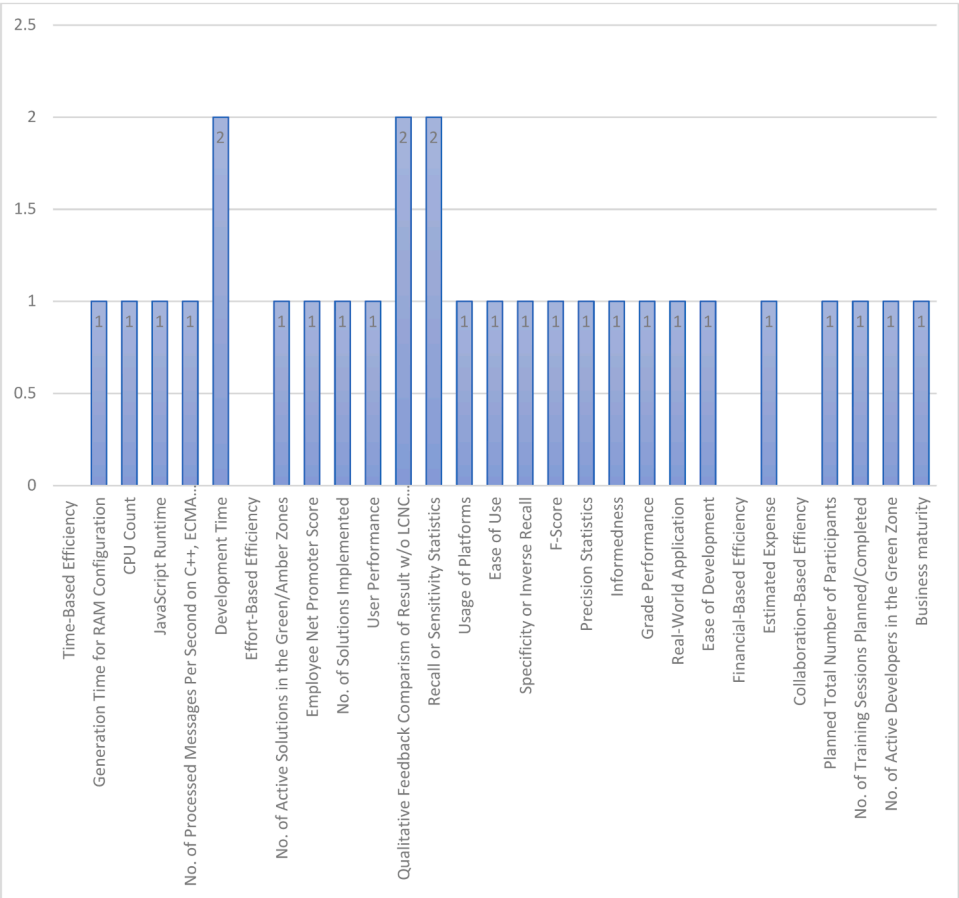


Fig. 6. Key performance indicators (evaluation measurements) are grouped according to the four major types of benefits identified (time-based, effort-based, financial-based, and collaboration-based efficiency).

deployment of LCN development technologies; it may also take some time before researchers have a firm grasp of applicable theories and methods to investigate the phenomenon. However, it is important to examine the theoretical and methodological lens by which researchers

have applied theories, frameworks, and methods to investigate it to date. Interestingly, it appears that most of the reviewed studies especially empirical studies do not ground their investigation of CD research on certain theoretical foundations.

However, a few studies have applied organizational theories to this topic, such as the resource-based view (RBV) (Iho et al., 2021), technology-organization-environment (TOE) framework (Hoogsteen and Borgman, 2022), organization implementation variability (OIVs) (Op 't Land et al., 2021), and theory of planned behaviour (TPB) (Thacker et al., 2020). Some of these theories have examined CD from individuals, groups, and organizational levels of analysis. Even without specific theoretical applications, other studies have developed frameworks to guide the process of ideas development on low-code development platforms (Krejci et al., 2021), support factors for CD adoption (Varajão, 2021), and empowerment of citizen developers (Wang and Wang, 2021). Additional studies include usability issues (Silva et al., 2021) and impact analysis framework (Overeem and Jansen, 2021) to provide theoretical perspectives on CD platforms.

The methodological approaches underpinning their various findings and claims have been discussed earlier to avoid repetition and are summarized in Fig. 3 below. Overall, researchers have employed a mix of qualitative and quantitative techniques, including practical observational and controlled module experiments and simulation techniques in this research area.

Moreover, some of the quantitative studies have generated data through online surveys, archival data such as blog posts from Stack

Overflow and Reddit, score rating experiments, process mapping, observational course assessments, etc. In addition, some qualitative studies have included interviews and audio/video experiments. While some studies have conducted only surveys or interviews, and a few have combined observation, interviews, and archival data. In reviewing the extant literature, we summarize by conceptualizing our understanding of CD adoption in the following section.

6. A framework on citizen development adoption for digital transformation

The framework (see Fig. 7) is derived from the SLR to explore the technological-driven benefits and challenges of adopting LCNC development platforms for driving DT initiatives in organizations. Our findings indicate that the benefits and challenges of CD adoption can be categorized into four aggregate themes: the nature of technology development platforms, management implementation strategy, employees' roles and digital skills, and performance evaluation metrics.

Firstly, the reviewed literature suggests the need to understand the nature of LCNC development technology platforms and whether they are fit, or unfit for use depending on the organizational motives, intents, or goals for adoption. Some issues of fit to be considered would be related

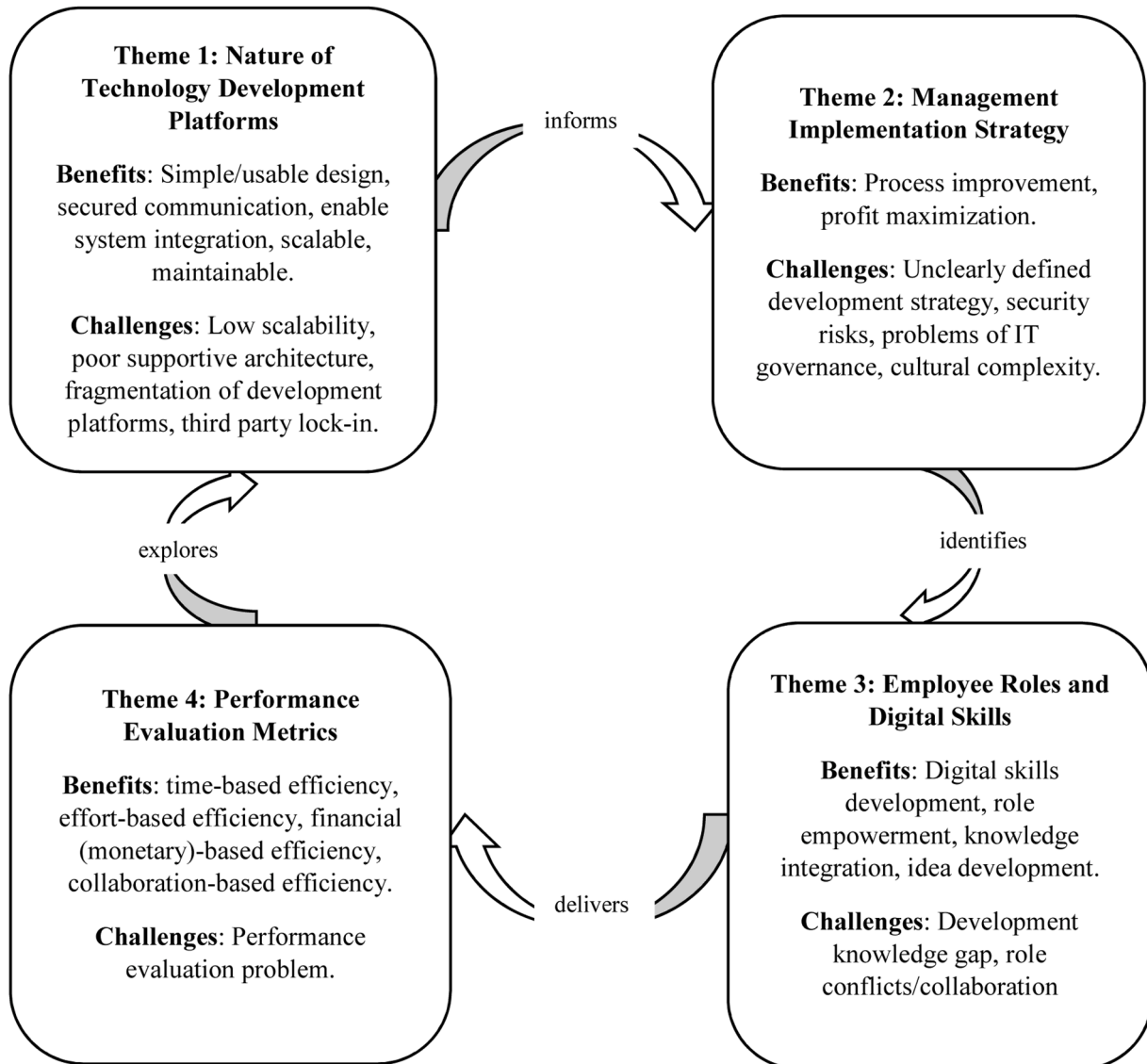


Fig. 7. A framework on the state of research in citizen development adoption for digital transformation.

to usability, scalability, level of customization and integration, security, maintenance, and third-party lock-in. [Alsaadi et al. \(2021\)](#) and [Lebens et al. \(2021\)](#) suggests the need for organizations to consider how selecting a particular LCNC technology will help provide the most and best solutions to their DT challenges (or business process used cases). This is why understanding comparative advantages and usability of different LCNCs and their suitability to specific business functions matter to an organization. In doing so, a number of authors have suggested that organizations should bear in mind how adopting LCNC technology solutions will fit into their business process and how it will address their challenges ([Bhattacharyya & Kumar \(2021\)](#)), as well as understanding the potential benefits to be derived and temporality of risks to be managed ([Hoogsteen and Boorgman, 2022](#)) due to the possibility of certain benefits becoming limitations for an organization ([Luo et al., 2021](#)).

Secondly, we feel that this 'litmus test' can then help organizations to think about the potential utility and suitability of such technologies for adoption, in order to come up with a strategy to implement the development process of their DT initiatives. Hence, setting clear goals and raising awareness by communicating across departments for adoption, stimulating a support culture, mobilising, and deploying employees for development, in addition to governing the general development process and controlling potential data security and privacy risks. As management implementation strategy is critical to achieving a successful LCNC and CD adoption, [Varajão \(2021\)](#) highlight a number of success factors, which include having a top management support, agile decision-making process and management, understanding of technology to be adopted and a strong commitment of the project team. In addition to managing the issues with governance and cultural change ([Prinz et al., 2022](#)).

Thirdly, however, part of the implementation process is to focus not just on the viability of adopting LCNC development platforms, but on the role of employees in driving its development. This is to take into consideration whether or not employees possess the development skills (see [Table 5](#)) and the need for training where necessary before and upon adoption in the DT journey ([Bernsteiner et al., 2022](#)). In this instance, [Krejci et al. \(2021\)](#) noted that understanding the role of individuals matters in adopting digital innovation in low-code development platforms. One of the reasons this is paramount is that it is possible that individuals may not have the capability to interact with certain LCNC platforms. As a result, there could be interaction conflicts due to knowledge gap for users ([Silva et al., 2021](#)). Thus, the need for organizational skill-enhancing interventions through training support would matter for their agility and adaptability in digital cultural environment towards leveraging, leading, and sustaining a successful digital transformation initiatives in organizations.

Finally, while there is limited evidence on ways to measure the benefits of LCNC and CD adoptions, there is some empirical evidence which indicates that organizations may need to examine time, effort, cost, and collaboration-based efficiency as key performance indicators. Depending on their selection of specific metrics pertinent to them, studies suggest conducting impact analysis on selected LCNC platforms whether the IT solutions developed on it can help the organization and its development teams achieve their business improvement goals ([Overeem and Jansen, 2021](#)). A feedback appraisal may also help in making decisions whether changes would need to be made to the LCNC systems, support tools, and the business processes. The cyclical nature of the framework essentially leads to a renewed cycle to meet new or evolving DT requirements and undergo new adoption stages.

7. Discussion: A research agenda

Citizen development is relatively a new phenomenon. This offers exciting opportunities to explore other areas worthy of research attention. While CD is one of the means to achieving a DT of business operations, our SLR has provided us with a better understanding of the benefits, challenges, and evaluation of CD adoption. We have also

considered some relevant questions posed for further research in the extant literature. Based on the literature highlights and synthesis according to our proposed framework for CD adoption, we suggest a research agenda that addresses (i) understanding the task-technology fit of LCNC development platforms on user job performance and innovation performance, (ii) lack of clarity on the characteristics of digital skills for CD and DT, (iii) the effect of role conflicts on network sustainable collaboration of developers, and (iv) insufficient and inconsistent CD performance metrics to evaluate CD success. Based on the reviewed studies and the framework developed to tease out the state-of-the-art about LCNC and CD practices, we feel it is important to group our research agenda within these four themes. To note in the following research agenda, and based on the systematic review, users in the context of CD adoption can be at the level of individuals and organizations.

7.1. Agenda I: understanding the task-technology fit of LCNC development platforms on user performance and innovation performance

The fragmented nature of LCNC technologies suggests that their features would not be the same across development platforms. In other words, different development platforms may have similar and dissimilar characteristics and advantages ([Ihirwe et al., 2020](#); [Kholiya et al., 2021](#); [Talesra and Nagaraja, 2021](#)) that may determine their fit for use in addressing different software development tasks for users. For example, studies have found interaction problems in learning a Precog low-code development platform based on users' knowledge and system requirements, which tends to have implications for achieving performance ([Silva et al., 2021](#)). In addition, while both Mendix and Salesforce provide options for system extensions to easily add modelling features and modify existing features, only Mendix can perform a multi-view modelling task, whereas only Salesforce can generate code for deployment support of business applications ([Ihirwe et al., 2020](#)). Therefore, the characteristics, perceptions of benefits, and utilisation of LCNC technologies may vary with different software development tasks ([Alsaadi et al., 2021](#); [Hoogsteen and Borgman, 2022](#); [Ihirwe et al., 2020](#); [Oltrogge et al., 2018](#)). While the literature suggests there is a need to continuously optimize the user-friendliness of software development platforms, additional research is required to explore how to harness the potential of AI in co-creating codeless technological platforms and the evolution of LCNC in developing and deploying more accessible, usable, and multi-dimensional business applications for organizations. This is because the integration and application of AI-driven support in co-creating codeless technological platforms will yield more faster, flexible, adaptable, maintainable, and learnable digital platforms for building highly customizable and scalable digital solutions for multi-purpose use by individuals and organizations of all sizes. [Table 8](#) highlights that a few studies have applied some notable theories to investigate CD adoption. These include the application of the theory of planned behaviour to understand students' perceived use of LCNC tools ([Thacker et al., 2020](#)); the use of resource-based view theory to understand how the use of low-code development platforms in terms of reusable development components, ease of mastery, visual interface and real-time editing, contributes to a user knowledge integration ([Iho et al., 2021](#)) and idea development ([Krejci et al., 2021](#)), respectively.

Future research can build on these contributions by investigating how task-technology-fit of LCNC development use affects user job performance and user innovation performance. In doing so, researchers can apply important technology acceptance theories like the Technology Acceptance Model (TAM) ([Davis, 1989](#)), Unified Theory of Acceptance and Use of Technology (UTAUT) ([Venkatesh et al., 2016](#)), Task-Technology-Fit (TTF) ([Goodhue and Thompson, 1995](#)) or Task-Technology-Identity-Fit ([El-Masri et al., 2022](#)) to contribute to research on CD adoption for DT. Although within the use of wearable smartwatch technologies, it is argued that rather than focusing on either actual task-technology fit (where both task and technology

characteristics align) or perceived task-technology fit, the latter studies suggest exploring them together on user utilisation and other performance-related outcomes. While also suggesting how it may shape a user's identity towards deriving a better satisfaction (El-Masri et al., 2022). Elshan et al. (2023) found that their interviews with 28 expert users of low-code platforms suggests that it permits greater involvement of customers in managing potential communication gaps in fulfilling development requirements, thus enabling them to derive higher client satisfaction. Future research should also explore the process by which organizations have adopted and normalized CD solutions. Normalized here refers to the organizational ability or capacity to integrate and sustain CD practices in work routinizations. In doing so, researchers may use Normalization Process Theory to understand how CD practices are embedded, enacted, and integrated into everyday organizational routines that affect performance (Carroll and Conboy, 2020; May and Finch, 2009). For example, the application of normalization process theory considers how both the top management and employees have shared understanding around the adoption of a new practice, such as adopting a new technology for improving a business practice. It considers how organizations mobilize organization-wide support among executives and employees, as well as pertinent resources (e.g., cost, skills, culture) to operationalize such a practice into everyday work routines. In addition to allowing for reviewing its impact on the business, people, and the entire organization for continued transformation and sustained performance. By considering this theory, researchers and practitioners can better understand the requirements, resource provisions, and maturity trajectory for LCNC and CD adoption.

7.2. Agenda II: Lack of clarity on the characteristics of digital skills for citizen development and digital transformation

In recent times, particularly since the time of COVID-19, the shortage of software developers has been attributed to the lack of digital skills. The term digital skills have gained broader appeal among researchers and practitioners and have been considered important for developing citizen developers, as well as in improving the skillsets of professional software developers. As digital transformation requires a new set of digital and software-related skills, the awareness of such functional capabilities to identify and develop them will be vital in meeting modern business needs, particularly in adopting more agile digital platforms like LCNC and CD solutions in organizations and society. However, there is no clarity as to what specific skills being related to the use of LCNC and CD practices are called digital skills in the extant literature. There can only be an educated guess of what digital skills entails and implies for reference by authors in some of these studies (see, for example, Metrôlho et al., 2020). Notwithstanding, the work of van Laar et al. (2017) described digital skills within the context of 21st century skills as general skills that enable people to operate ICT related tools, devices, and systems in a knowledge-based society and digitalized world. These skills include technical, information management, communication, collaboration, creativity, critical thinking, ethical awareness, cultural awareness, flexibility, self-direction and related lifelong learning skills. While problem-solving skills remain critical to software developers, and digital skills can help us to use ICT tools correctly, Wing (2006) suggest that computational thinking enables everyone to solve problems, design systems and understand human behaviour, which goes beyond fulfilling technical requirements for software development activities. Although within the software development context, a number of specific skills have been proposed which we have classified as technical and soft skills (see Table 5). For example, technical skills may be required to understand the development of business software applications. These include data mastery of information systems and technology solutions (Bernsteiner et al., 2022; Metrôlho et al., 2020), designing and creating intuitive applications (Lara-Prieto and Flores-Garza, 2021), using sensors and low-code interfaces (Colantoni et al., 2020), coding (Alsaadi et al., 2021), using operational technologies (Lara-Prieto and

Flores-Garza, 2021), handling databases, and understanding development environment and deployment processes (Metrôlho et al., 2020), among others. While soft skills entail the ability to comprehend problem scenarios (Bernsteiner et al., 2022; Metrôlho et al., 2020), reason creatively and logically (Wang and Wang, 2021), plan to implement and communicate technology solutions (Elbatanony and Succì, 2021; Lara-Prieto and Flores-Garza, 2021) and time management of software development processes (Metrôlho et al., 2020). However, our SLR revealed that there is more research on technical skills than soft skills.

Future studies can build on our findings by examining what type of technical skills and soft skills are made up of digital skills that are relevant for citizen developers. For example, researchers have emphasized the need to look for technical skills (knowledge of 'technical standards, software development and configuration'), business skills ('knowledge of general business processes, practices, procedures') and administrative skills ('timely preparation and maintenance of project documentation') (Aladwani, 2002, p. 205) in individuals for software development. Our findings also suggest the need for intensive skills development in science technology engineering and mathematics (STEM) related educational pedagogy where some of these skills should be taught to students in raising future developers as a continuous response to skills gap in software industry.

7.3. Agenda III: The effect of role conflicts on network sustainable collaboration between citizen developers and professional software developers

We discussed earlier that software development requires work collaboration between business professionals and IT professionals (Bexiga et al., 2020). However, as LCNC technologies provide a means for knowledge integration and idea development in rapidly replacing technology knowledge or technical knowledge of software developers (Iho et al., 2021; Krejci et al., 2021); the tendency to remain as the "go-to" persons for technical expert knowledge diminishes. Hence, role conflicts can arise which can make it harder to manage effective collaboration between them (Al Alamin et al., 2021; Lebens et al., 2021; Wang et al., 2021). This can be exacerbated when professional software developers challenge citizen developers over shadow IT problems or security issues in developing business applications outside their expert advice (Elshan et al., 2023; Hoogsteen and Borgman, 2022; Oltrogge et al., 2018). This begs the question of how and when the empowerment of low-code technologies may have a negative bearing on sustainable collaboration (Luo et al., 2021). In that regard, researchers should investigate the negative effects of role conflicts on for example, agreeableness, openness to experience (personality-related), trust, emotional reaction, anxiety, dissatisfaction (cognitive-related), task coordination, communication, knowledge exchange, and interpersonal learning (social-related) (Mathieu et al., 2008) between citizen developers and professional software developers. Studies on social network mechanisms have found that a relationship that is embedded with business, technical and friendly relations between two individuals tends to enjoy more sustainable collaboration than otherwise (Ajimati et al., 2021). In other words, a lack of or limited interactions can lead to a communication gap for sustained collaboration between business and professional developers. Nevertheless, research has that "employees are willing to seek advice from those with whom there is task conflict despite the simultaneous presence of relationship conflict (e.g., friendships), those characterized by multiplex conflicts provide greater access to others with divergent perspectives while also reducing the need for advice seekers to invest resources towards protection and maintenance of the positive portion of the relationship." (Marineau et al., 2018, p.595). All these factors can adversely affect sustainable collaboration or work collaboration efficiency between citizen developers and professional software developers in organizations.

7.4. Agenda IV: Insufficient and inconsistent citizen development performance metrics to evaluate citizen development success

Upon CD adoption, understanding how organizations evaluate the successful or failed transformation of business processes can be valuable for research and practice. The SLR revealed that CD performance can be observed based on its capacity to reduce development time, effort, financial and degree of collaboration between developers during the software development process (Bexiga et al., 2020; Pacheco et al., 2021; Pantelimon et al., 2019). While existing studies have measured CD time and effort using coding parameters (Ali et al., 2021; Pantelimon et al., 2019), the level of accuracy or precision of software behaviour, and recall rate of components covered (Pacheco et al., 2021; Silva et al., 2021), comparing development time using low-code and non-low code approaches (Bexiga et al., 2020; Pichidienthum et al., 2021); there is still insufficient understanding about how these can be more measured quantitatively rather than qualitatively more generally. There is also a lack of consistency concerning how time – and effort-based efficiency were measured. In addition, there are no real measures through which financial, and collaboration were also measured (see Table 8). Overall, there is limited empirical evidence of key performance indicators of CD effectiveness in organizations. Extant literature has shown that systems development effectiveness can be measured based on achieving business needs, expected quality, fulfilling functional requirements and overall project objectives. While systems development efficiency can be measured by estimated cost, schedule, development time and programming effort (Tiwana, 2012). There is an urgent need for more empirical studies on how practitioners have measured CD in organizations (Carroll et al., 2023). It would also be worthwhile for researchers to understand whether a set of key performance indicators tend to vary or not with different development platforms for different DT initiatives depending on the organization DT maturity. For example, Fig. 6 suggests that there is no consensus on valid KPIs that have been used to evaluate CD and LCNC benefits across the empirical studies.

Our study explains the role in which CD plays in driving the DT journey in organizations, there is a lack of empirical and theoretical evidence on the maturity ladder of CD lifecycle management in the extant literature (Prinz et al., 2022). Although CD is still a relatively new area for researchers and practitioners because of the availability of small used cases, hence information about its maturity level may be unavailable to researchers. Nonetheless, researchers should consider understanding the maturity level of CD adoption among practitioners in different organizations by taking a leaf from the DT maturity level or something similar for theoretical and managerial insights. In doing so, it can help us to know and match essential requirements for business transformation decisions in the pre-adoption, adoption, and post-adoption phases of CD practices (Carroll et al. 2024).

8. Conclusion and limitations

The goal of this research was to conduct an SLR to understand the benefits, challenges, and performance evaluation of CD adoption for DT in organizations. The key research questions were to understand (RQ1) “what are the benefits and challenges associated with the adoption of LCNC technologies and CD practices in organizations”, and (RQ2) “how do organizations evaluate the methods they use in measuring the benefits of LCNC adoption and CD practices?” The answers to our research questions may be found in Section 4, where we positioned our findings into four aggregate themes, namely, the nature of LCNC development technology platforms, management implementation strategy, employees’ roles and digital skills, and performance evaluation metrics. These findings were represented as a framework in Section 6. In answering our RQ1, on the nature of LCNC development technology platforms: our findings suggest that the benefits of CD offer adopters the ability or capacity to use LCNC platforms to easily design and develop simple, usable, customizable, scalable, and maintainable software and

mobile applications. Other benefits include its tendency to enable secured communication and system integration with external services. On the other hand, our findings suggest that adopters still face some challenges such as low scalability, poor supportive infrastructure, fragmentation of development platforms, and third-party lock-in. Regarding management implementation strategy, our findings indicate that organizations leverage LCNC development platforms to improve their business processes by driving operational efficiency in time, effort, finance (monetary cost), and collaboration to maximize profit. However, our findings further reveal that organizations still face the problem of determining how best to evaluate CD performance. Regarding employees’ roles and digital skills, our findings reveal that CD adoption is a strategic response to the shortage of professional software developers. In doing so, this has contributed to the technological motivation for employing and deploying LCNC development platforms in building software applications to solve workflow problems. Specifically, the benefits of CD adoption enable employees to gain technical and soft-related digital skills for developing software applications. In doing so, employees become empowered within their work roles to integrate knowledge and develop ideas toward innovation. However, the adoption of CD is still challenged by the gap in the development knowledge as well as the problem of role conflicts that threaten sustainable collaboration between citizen developers and professional software developers.

Overall, our answers to the RQ1 revealed that CD and LCNC benefits contributes to employees’ digital development efforts and enhances their motivation to freely engage in organizational wide digital transformation. Thus, promoting collective intelligence among the organizational workforce, especially business professionals, in engaging and participating in problems-solving and innovative activities for business solutions. Digital literacy is also needed to operate in this digital era. In addition, the simplicity and user-friendly design of LCNC platforms help organizations to reduce their cost of hiring and training professional developers in driving software solutions to business operational problems in organizations. In terms of its challenges, we found that organizations still struggle with strategizing how best to tap its potential benefits in enabling operational performance. A key aspect is deciding the need and how best to define acceptable key performance indicators (KPIs) in evaluating the return of investment (ROI) on technologies and employees in organizations. The possibility of being locked-in by platform vendors may also result in incurring higher switching costs for adopters, in case there is maintainability or longevity issues for providers.

Furthermore, in answering our RQ2, while our results revealed some clear evidence around the measurable benefits of CD and LCNC, for example, in reducing development time and estimated expense; our results suggest that the different properties pertaining to different platforms and use cases poses challenge for organizations in determining their evaluation criteria for assessing the benefits of CD and LCNC. In addition, in comparing our SLR to the work of Bucaioni et al. (2022); Berardi et al. (2023) and Sufi (2023), our work extends their multi-vocal systematic review because we were able to identify and provide comprehensive review of the benefits, as well as practitioner perspectives on the evaluation of the performance benefits of CD solutions within LCNC platforms. Notwithstanding the fact that our research was not focused on model-driven engineering.

This research is not without limitations. We have focused our SLR on low-code and no-code associated terms within CD perspectives, which has informed our understanding of LCNC research studies and current findings. However, this study does not cover research thinking linking this to model-driven engineering to maintain focus on exploring our research questions for this SLR. This might have limited our search terms to advance our understanding of the benefits and challenges of CD within the broader context of the literature in computer science and engineering fields. While our proposed framework on CD is informative, it does not offer further specific guidance on how this can be implemented by organizations beyond the boundary of the literature

examined. It also needs to be further explored by researchers for more contributions to the LCNC and CD research field.

Overall, our current work contributes to understanding LCNC technologies and CD practices for adoption and their implications for researchers and practitioners. Our framework suggests that organizations would need to explore LCNC technologies for suitability in terms of inherent benefits and challenges before being considered for development in organizations. This is expected to inform organizational management decisions on devising a clear strategy for developing and implementing such LCNC technologies and CD practices as business solutions in organizations. More importantly, such a strategy requires identifying the different roles of employees and their digital skills that will enable them to match technology development tasks to deliver expected performance. Lastly, the need for ongoing review of employees' performance depends on how they are evaluated based on organizational-specific key performance indicators. This would further dictate prolonged use and optimization of such LCNC technologies for continuous learning and adaptability to changes in further DT requirements, and for more improved business operations in successfully sustaining DT efforts. Conclusively, each theme within our framework opens opportunities for how the adoption of LCNC technologies and CD practices may be approached by organizations. Additionally, for researchers, our SLR serves as avenues to drive new frontiers in IS research related to DT and LCNC digital innovation (e.g., design, collaboration, knowledge sharing, socio-technical issues, etc.) for citizen developers in

organizations.

CRedit authorship contribution statement

Matthew Oladeji Ajimati: Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Noel Carroll:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Mary Maher:** Writing – original draft, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 8

Overview of The 40 Primary Studies Highlighting Their Theories, Methods, Benefits/Findings, and Performance Evaluations.

Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
1			Education	Online survey: Online studio classroom. 50 Design students.	The use of LCNC ecosystem for design education can reduce the time and complexity (i., remove programming barrier) to synthesize UX design process for interactive design learning.		Abidin et al. (2021)
2				Data dump in online developer forum: 5000 Stack Overflow Posts (questions/ accepted answers)	Low-code software development challenges are related to customization, platform adaptation, database management and third-party integration in the development phase, in addition to automated testing.		Al Alamin et al., (2021)
3				Data dump in online developer forum: 33,700 Stack Overflow Posts (questions + answers), 60 SO Tags 38 LCSD platforms.	Challenges relate to applications customization, database and file management, platform maintenance, and third-party API integration.		Al Alamin et al., 2023
4				Survey 49 Professional programmers/ developers: 19 Users of LCDP 30 Non-users of LCDP	LCDP reduces application development time but with limited level of scalability.		Alsaadi et al. (2021)

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Table 8 (continued)

Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
5	Metamask		Supply chain management	Survey 20 Industrialists No user study	No-code development platform reduces development time and investment in building customized supply chain management applications, including migration of existing SCM systems to blockchain.	Measured by (1) the time consumed for developing a smart contract and (2) an estimated expense. Measurement indicators include “efficiency, user performance, ease of use, and real-word application	Asawa et al. (2021)
6	PowerApps, Oauth2		Education	Score rating experiment 3 Swedish universities 2 Lectures Students (No specific sample)	By leveraging citizen development practice for a digital campus in learning course, university can scale up their digital transformation process.		Baumgarten et al. (2020)
7	UMUX, MUM-Lite		Education	Survey 3 Slovenian business organizations (IT, banking and finance, healthcare and pharmacy, insurance, and energy sector) 78 Employees 35 Master students in IT	The use, adoption, and usability study of LCNC among Slovenian employees and students suggests a low level of adoption due to the issues of development feasibility, lack of knowledge, third-party lock-in, and measurement problems.		Beranic et al. (2020)
8			Software development	Interviews: Semi-structured interviews. 5 Professional software developers 5 Citizen developers	Identifies and differentiate specific requirement skills (i.e., soft/technical) between citizen developers and professional developers.		Bernsteiner et al., 2022
9	OutSystems		Software engineering	Process mapping, model-based methodology Interviews: 25 OutSystems' professionals (Developers, designers, project managers, tech leads, team leads, people allocation managers).	Using low-code technology to transform UX/UI design artefacts saved between 20 and 75% of complexity and effort of development teams.	Qualitatively measured by the Front-End Experts team comparing results with and without (past work) using low-code platforms.	Bexiga et al. (2020)
10		Built on Diffusion of Innovations (DOI) and Technological Organizational Environment (TOE) theory frameworks	Digitization of supply chain management	A semi-structured open-ended questionnaire, In-depth interviews 20 Subject-matter experts in digital transformation, procurement, and strategy in supply chain management functions.	LCNC application platforms reduce organizational dependency on coders and cost of operations, streamline processes for vendor and supplier management, including easy maintenance.		Bhattacharyya & Kumar (2021)
11				Interviews: 18 Semi-structure interviews with senior managers/users	Uncovered the steps (i. e. sensemaking, stakeholder participation, collective action, evaluating progress) by which Shell normalized citizen development.	Shell's DIY development KPIs: Training sessions/ events (e.g., Bootcamps): completed, planned, total number of participants, business maturity; usage of power platforms (no. of active developers in the green zone, no. of active solutions in the green and amber zones, employee net promoter	Carroll & Maher (2023)

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Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
12	OutSystems	Technology Organization Environment (TOE) framework	Education: Scalability in project-based learning in software engineering course on digital skill development	Interviews: 39 Domain experts in using LCNC.	Examined the utility of LCNC in developing smart personal assistance applications.	score; deployment (no. of solutions – implemented, in development, total; total value (\$ millions), implemented, in the pipeline, tracked by lines of business; operations (no. of DIY coaches, no. of certifications (DIY passports/DIY Visas/ eternal qualifications), no. of apps on the diyXchange platform.	E. Elshan et al. (2023)
13				Interviews: Semi-structured interviews 28 expert users	Explored the benefits (e.g., digitization/modernization) and challenges (e.g., shadow IT) of low-code platforms		E. Elshan et al. (2023)
14				Controlled modelling environment for software engineering course, Survey 161 Students	Low-code platform enable the realization of scalability in project-based learning.	Measured by (1) grades obtained (progress/ learning achievement) (2) feedback (to assess individual/group performance)	Fernandes et al. (2020)
15				Survey 6 Case studies from different sector 12 Participants (IT developer, IT manager, IT consultants, hospitality manager, business analyst, global product owner, tech solutions director, senior factory operator (non-IT), product owners.	Citizen development adoption decisions are influenced by “multiple risk perceptions, active top management support, project-based business-IT alignment, centralized IT governance, and business network systems.”		Hoogsteen & Borgman (2022)
16	Airtable, AppSheet, Betty Blocks, Caspio Bridge, FileMaker, Kintone, KISS Flow, Quick Base, Tibco Cloud Live Apps, TrackVIA, Viyl	Resource-based View (RBV)	Knowledge integration, innovation	10 Interviews, product reviews (including executives/employees) 201 Reviews from the selected 11 LCDPs.	LCDP as knowledge integration mechanism for business and IT alignment in supporting innovation development.		Iho et al. (2021)
17	IFTTT, Zapier			Citizen developers Software engineers	Low-code development environment support the ability of citizen developers in the development and deployment of management workflows.		Indamusta (2021)
18			Intelligent automation services	Archival reports: internal documentation, Interviews: 2 participants at management level			Kedziora et al. (2023)
19			The thing description and Internet of Things	Experiment 3 Case studies 20 Consumers	-code shadow things enable the deployment of scalable, reliable, safe, and feasible existing IOT systems with no programming effort of developers	Number of time it takes to generate Exposed Things for different RAM configurations and CPU count	Korkan et al. (2020)

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Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
20	Airtable, AppSheet, Betty Blocks, Caspio, FileMaker, Kinton, Kissflow, QuickBase, Scopeland, TIBCO, TrackVia and Zudy.	Organization implementation variability (OIV)	Idea management, Digital innovation	10 Expert interviews/ 114 pages transcript on LCDPs (executives, employees: CEO, COO, VP of sales, senior developer, lead developer, platform developers, project manager, account manager, product manager) Archival data: Analyst reports, Vendors' documentation (56 documents/509 pages) User reviews on LCDPs (953 reviews/667 pages). 12 selected LCDPs	LCDPs empower employees with flexibility and autonomy to facilitate idea development for digital innovation.		Krejci et al. (2021)
21				Survey: 38 IT Professionals from LinkedIn, Twitter/ New IT Conference	LCNC enable organizations of all sizes with workflow automation toolsets for developing technology solutions to business problems/shortage of software developers.		Lebens et al. (2021)
22				DEMO/OIV experiment on proof of concept 4 IT professionals: DEMO (network) expert, junior business analyst, senior developer, junior developer	Promotes the flexibility and autonomy for human actors to responsibly bypass reprogramming of software for process automation (OIVs) based on RUN-time modelling of products and services.		Op 't Land et al. (2021)
23				Online repository: Q&A-based data. 301 LCD posts on Stack Overflow/Reddit	LCD provide learnable and easy to use drag and drop graphical interface for process/workflow automation without coding. Concerns are associated with access problem to source code, vendor lock-in and project solution fit.		Luo et al. (2021)
24	Microsoft Power Apps		Education	Surveys: A pre-training survey: 18 respondents A post-training survey: 8 respondents Interviews: Semi-structured interviews: 4 participants	Examine the potential impact of LCNC technologies (e.g. in facilitating technical skills) toward educating computer science students in secondary schools.		McHugh et al., 2023
25	OutSystems		Education	Survey: 21 Trainees with a science technology engineering and mathematics.	Low-code platform encourage retraining of professionals from (STEM) with the necessary IT skills to fill gap in the global shortage of software developers.		Metrolho et al. (2019)
26	OutSystems		Education	Survey: 19 Trainees with a science technology engineering and mathematics.	Enterprise-Higer Education collaborative using LCDPs as strategic mechanisms for reskilling of professionals with STEM technical skills and training of		Metrólho et al. (2020)

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Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
27	DisCo Par		Smart agriculture applications (SAAs)	Experiment design	unemployed people to utilizing ICT skills using its high productivity visual development, database scripts, ultra-fast visual modelling, and on-device business logic and access features of LCDPs. Low-code development tools offer opportunity for farmers to develop smart agriculture applications for timely notifications of farm conditions.		Oteyo et al. (2021)
28	AFAS		Software evolution and impact analysis	Observations Citizen developers LCDP developers Operations engineers No sample	Impact analysis as foundations to understand LCDPs evolution and utility.		Overeem & Jansen (2021)
29	Mendix, Microsoft Power Apps			Interviews: Semi-structured interviews 7 Organizations 16 Executives and technologists	Examined steps for organizations to take in adopting LCD		Novales & Mancha (2023)
30			Development of customer-facing applications by UX/UI designers	Interviews: 2 Front-end experts 3 Designers	Low-code platform improve the collaboration efficiency (effort) between designers and front-end developers in producing greater number of pages and screens delivered to customers.	Implementation was measured by (1) calculating the <i>Precision</i> and (2) <i>Recall</i> statistics. <i>Precision</i> relates to the tool's ability to identify and instantiate a component correctly and <i>Recall</i> relates to how many components in the designs our tool covers	Pacheco et al. (2021)
31	SDL4Snap!		Education	Observational course assessment Industry 4.0 postgraduate students No sample mentioned	Low-code approach facilitates student engagement and learning outcomes in developing software applications compared with non-low-code programming models.		Palomés et al. (2021)
32				Platform users' study. No sample size mentioned.	Low-code approach enable seamless end-to-end integration and management between heterogeneous hardware sensors (devices) and IOT platforms. Less programming effort: ease of coding, flexibility of loading and execution, good documentation for JavaScript and JavaScript interpreters	Measured by the number of processed messages per second based on C++, ECMAScript 5, and JavaScript runtime (best in terms of ease of development/performance)	Pantelimon et al. (2019)
33	Odoo		Low-code development platform for software development in enterprise systems	Lab experiment, observation 15 Users 3 Developers	Low-code development concept for automatic code generation and module creation reduce development and build time by 20%	Timing method: This was measured by (1) comparing the build time using the normal method and (2) the module generator (i.e., low-code approach).	Pichidienthum et al. (2021)
34				15 Expert interviews 4 Practical studies on LCDP challenges	The challenges of LCDPs for scientific and practitioner communities include knowledge transfer,		Prinz et al. (2022)

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Number	LCNC Platforms	Theory	Context	Methods	Benefits/Findings	Performance Evaluations	References
35				Interviews: 5 Case companies 5 Interviews	perceived differences in social and scientific contexts, including issues of "governance, culture change, and evaluation of LCDPs". Explores the requirements for LCDPs, benefits, challenges and limitations.		Schenkenfelder et al. (2023)
36	PRECOC		Software development: Using descriptive cognitive model to determine a low-code development platform's system usability	Audio/Video experiment 20 Participating users (10 experts, 10 novices)	The extent of users' knowledge application determines usability issues in performing low-code development tasks. These include over-decomposition, under-decomposition and no correspondence conflicts among users.	Users' interaction performance was measured by (1) Sensitivity or recall (2) specificity or inverse recall (3) Accuracy (4) F-Score and (5) Informedness	Silva et al. (2021)
37	Zapier System	Theory of planned Behaviour (TPB)	Education	Survey 500 Students from business management, marketing, accounting, economics, finance, CIS, entrepreneurship.	Students perceive themselves as technology users of low-code tools not developers but open to learning its utilization for work automation.		Thacker et al. (2021)
38			Software engineering	Observations, 7 interviews: general manager, operation manager, system vendor, R&D team leader	Industrial domain experts are more inclined to utilize low-code platforms to develop industrial applications with less collaboration with professional software engineers.		Wang et al. (2021)
39	Microsoft Power Apps		Education	Controlled module teaching experiment: 17 students	Students find low-code development projects to be useful for problem-solving, creativity, self-regulation, and teamwork activities.		Wang & Wang (2021)
40	RESTsec			Scenario simulation Designers Users	Low-code platform enable developer to model, integrate, and create seamlessly secured enterprise services for automation.		Zolotas et al. (2018)

Note: Please note that blank cells indicate that data was not available for reporting at the time.

Data availability

No data was used for the research described in the article.

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